

Designing Size Inclusive Fashion Assortments

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Problem Definition: Assortment optimization methods help fashion retailers improve profitability and customer satisfaction. By focusing primarily on revenue maximization, these methods often overlook size inclusivity, leading to the under-representation of plus-sized customers. This raises fairness concerns and can negatively impact brand perception. **Methodology:** We develop a choice model that allows for size substitution and introduce a measure of pairwise fairness among customer size groups. We integrate these into an online fashion retailer’s in-season assortment optimization problem that allows the retailer to randomize across multiple displayed assortments. Using a selection-based formulation, we transform the combinatorial assortment problem into a polynomial-size linear program and develop a backward mapping that recovers an exact optimal randomized assortment policy. We extend the framework to two operational settings: size filtering, where customers select their size before browsing options, and option substitution, where customers may return to the category page after encountering an unavailable size. We estimate size-substitution behavior using data from our industry partner and use the estimates to calibrate extensive numerical experiments. **Managerial Implications:** Our simulations show that meaningful reductions in size-based disparity can be achieved at modest revenue cost, whereas aggressive equalization becomes increasingly expensive and may require worsening the experience of better-served size groups. Fairer displays do not require wholesale changes to the commercial assortment; they are implemented mainly through selective pruning and limited rotation among complementary assortments while retaining most revenue-maximizing options. Size filtering can improve both revenue and fairness, whereas allowing customers to revisit the category page after encountering an unavailable size operates primarily as a revenue-recovery mechanism. Because the effect of a given fairness weight depends on category characteristics, retailers should evaluate their own revenue–fairness frontier rather than rely on a universal fairness weight.

Key words: assortment optimization, size inclusivity, choice model, pairwise fairness, randomization

1. Introduction

Fashion retailers utilize assortment optimization methods to enhance profitability, reduce excess inventory, and boost customer satisfaction (*Style Arcade* 2024). Despite their financial success, these methods do not address *size inclusivity*, and plus-sized customers experience significantly higher unavailability rates. For example, an analysis of the top ten womenswear retailers in the UK by *Retail-Week* (2022) revealed that the availability rate for size 20 is just 53%, compared

to 76% for size 10. This disparity in size availability across colors and styles is a “subtle form of social discrimination,” negatively impacting customers’ brand perception (Volonté 2021). Given that perceptions of a brand’s diversity and inclusivity efforts affect purchase decisions of 80% of customers (Kantar 2025), the absence of size inclusivity is surprising.

Size non-inclusivity is not a recent phenomenon, but a legacy of decades-old sizing standards (The Interline 2025), and remains an industry-wide challenge today (Business Insider 2025). For instance, although 67% of American women wear size 14 or above, many styles are still not offered in these sizes (Time 2016). Importantly, unequal size availability need not reflect an explicit intention to disadvantage particular size groups. It can emerge through a sequence of otherwise standard operational decisions. Extended-size garments cannot always be produced by simply scaling up straight-size designs because body shape varies with size, often requiring additional pattern development and fitting (Self Magazine 2018, Vogue 2022). As a result, some styles may never be offered in extended sizes. Among styles that do carry extended sizes, differences in production costs, demand estimation, and inventory planning can create differences in inventory depth across sizes. Larger garments require more fabric and can involve greater development costs (Vogue 2022, Business Insider 2025), while size curves estimated from historical sales can understate demand for sizes that became unavailable earlier (Toolio 2026). Standardized pre-packs can introduce additional mismatch because a limited set of size-ratio templates need not closely match the size distribution of every location. Once the selling season begins, heterogeneous demand and limited replenishment can further transform these initial differences into persistent broken size runs.

We do not seek to identify which of these mechanisms is responsible for a particular observed disparity, nor to estimate their relative importance. Rather, they illustrate how a retailer can arrive partway through the selling season with asymmetric size availability even in the absence of an explicit policy of excluding particular size groups. We define an *option* as a style and color combination and an *SKU* as a particular size of that option. Once the season is underway, the retailer largely inherits which SKUs are available: an assortment decision cannot restore a missing size. It can, however, control which options customers see and, consequently, how frequently different size groups are exposed to the stockouts embedded in that inventory.

A seemingly straightforward response is to remove every option missing one or more sizes and display only options available in all sizes. Although this prevents customers from encountering broken size runs, it also hides options with sellable inventory in a subset of sizes, potentially reducing full-price sales and increasing the inventory that must later be marked down, cleared through secondary channels, or sent to landfill. The retailer therefore faces a distinct in-season decision problem. We develop a model to answer the following question: *How can online fashion*

retailers optimize their displayed assortment to maximize revenue while mitigating concerns about unfair size representation?

Our study is driven by a collaboration with one of Europe’s largest online fashion retailers. Like many in the industry, our partner faces the challenge of introducing a wide range of new products to meet rapidly evolving customer preferences. During a typical spring/summer season, the retailer offers around 30,000 options across categories. Table 1 shows the percentage of customers and the mid-season unavailability rate for different sizes in the women’s shirts category in a data sample from our industry partner. Although anecdotal and based on censored sales data, Table 1 provides confirmatory evidence on the *systematic disparity* that motivates our work: while size groups XXL and L account for the same share of customers, the unavailability rate for size group XXL is more than twice that of size group L. Importantly, such disparities need not reflect deliberate discrimination or managerial neglect; they can arise as an operational consequence of heterogeneous demand and inventory conditions.

A shift in our industry partner’s strategy further illustrates why the in-season display problem can remain relevant even after upstream size coverage improves. Initially, the retailer largely excluded size XXL from its production line: only 0.2% of the women’s shirt SKUs introduced into the online inventory were of size XXL. After the retailer changed its strategy, this share increased to 28%. Yet, as Table 1 shows, plus-sized customers continued to face substantially higher levels of mid-season size unavailability. We do not interpret this observation as evidence about the cause of those stockouts. Rather, it illustrates that improving initial size coverage does not necessarily eliminate asymmetric availability as inventory evolves over the season.

Size	XS	S	M	L	XL	XXL
Unavailability Rate	22.2%	19.5%	20.4%	29.8%	32.2%	68.7%
Customer Distribution	10%	20%	22%	16%	16%	16%

Table 1 Customer size distribution and mid-season size unavailability rates in the women’s shirts category.

Using this as our main setting, we take inventory availability as exogenous and study the retailer’s in-season assortment decision. Most SKUs managed by fast-fashion retailers (more than 75% at our partner retailer) fit this setting: once the initial shipment reaches the central warehouse for the online store, no replenishment occurs from suppliers. Thus, after the season begins, availability is effectively fixed, and the retailer’s role shifts from pre-season assortment decisions (often called category planning) to deciding which options to display. We modify this in-season assortment problem by incorporating a fairness notion based on disparities in customers’ experienced size unavailability.

Our approach is therefore complementary to, rather than a substitute for, better upstream planning. Design, size-curve estimation, purchasing, allocation, and replenishment determine how

much size scarcity and how many broken size runs the retailer inherits; the assortment display determines how customers are exposed to the consequences of that scarcity once it exists. No assortment optimization method can manufacture a missing SKU. Nonetheless, even with inventory fixed, the retailer must still decide whether and how to display options whose size configurations are incomplete. To the best of our knowledge, this is the first paper in the assortment optimization literature to make size inclusivity its core objective.

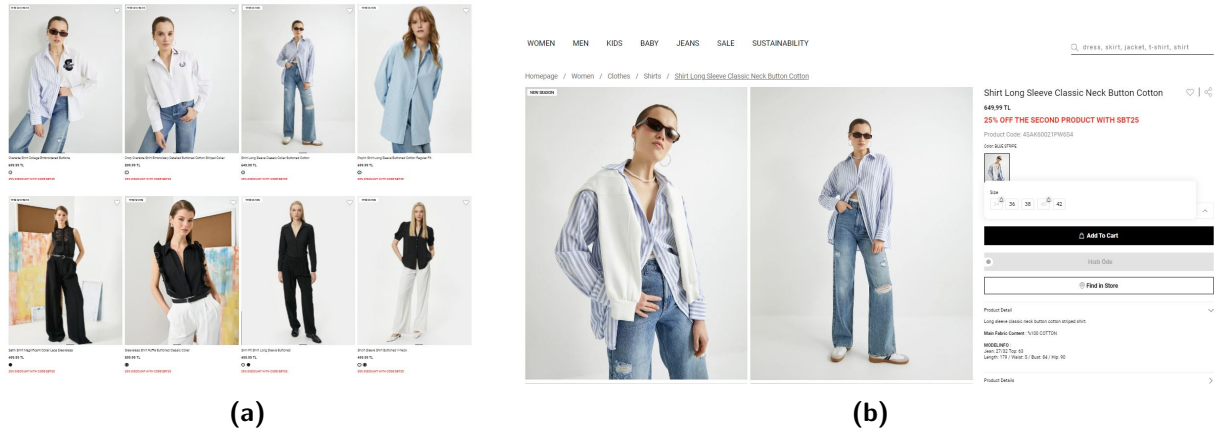


Figure 1 Figure (a): The category page for women's shirts. Figure (b): A sample option's page.

To explain our main contributions, we first review the problem setup and terminology. The sale of an SKU from option j generates revenue r_j for the retailer. The inventory has n distinct options, and the retailer decides the option assortment $S \subseteq \{1, \dots, n\}$. An option may not be available in all sizes, so we denote the size configuration of option j by $C_j \subseteq G$ where $G = \{1, \dots, g\}$ is the set of SKU sizes. We let $\mathcal{C} := \{C_j | j \in S\}$.

In the baseline model, a customer first visits a product category page similar to Figure 1a, considers all options in that category, and chooses option j according to a Multinomial Logit (MNL) model. After clicking on option j , she lands on an option page similar to Figure 1b. After observing C_j , she may choose an SKU from option j or leave the store. If her preferred size is unavailable for option j , she may substitute it with another size of the same option. We categorize customers into g size groups based on the SKU size that fits them best and define ‘unavailability’ as the probability that a customer in a given size group experiences size unavailability. This allows us to develop a *pairwise fairness* metric that measures the gap between the highest and lowest unavailability levels across size groups. (Our methods also apply to *Rawlsian max-min fairness* metrics.) Note that our use of the term fairness concerns disparities in service outcomes rather than the retailer’s intent. We set the objective as expected revenue minus the fairness metric multiplied by λ , where λ represents the *price of unfairness*. We next summarize our main results and contributions.

First, we prove that when all sizes are available for all options, the optimal option assortment has the same revenue-ordered solution developed in Talluri and Van Ryzin (2004) and, as such,

is unaffected by fairness concerns. However, full size availability is rarely realistic. During most of the selling season, retailers operate with a broken assortment. Displaying such options potentially causes fairness concerns because some size groups may systematically face less attractive assortments than others. Hence, the optimal display policy depends on the fairness weight λ . We develop the *fairness-aware* static assortment problem for online retailers, who can randomize across multiple assortments rather than display a single fixed assortment (Wang et al. 2024).

Second, we employ and extend the *selection-based* method (Gallego et al. 2015) to offer an exact solution to the problem. Because there are 2^n feasible assortments, the retailer’s problem has exponentially many decision variables. We therefore first transform it into a linear program (LP) in which, instead of choosing a display probability for each assortment, the retailer chooses a display probability for each option. This LP has $\mathcal{O}(n)$ decision variables and constraints, allowing it to be solved quickly by commercial solvers. We then transform the optimal LP solution back into assortment display probabilities. In addition to being optimal, this solution uses at most g^2 distinct assortments, supporting its practical implementation.

Third, we extend the problem formulation and solution method to characterize how additional operational levers can help the retailer to improve revenue, fairness, or both. These extensions include allowing customers to filter by size on the category page before choosing an option (i.e., size filtering), and encouraging customers who substitute across options to revisit the category page when their preferred size is unavailable.

Finally, we use a granular dataset provided by our industry partner to estimate the size-substitution coefficient, applying the difference-in-differences method of Li et al. (2023). The treatment group consists of SKUs whose neighboring size is unavailable, while the control group consists of SKUs from the same option that are unaffected by the unavailability of a neighboring size. The estimates (available in Appendix EC.6) imply a size-substitution coefficient for each neighboring size of approximately 25%, close to that reported by Li et al. (2023). We use this estimate, together with other features of the industry data, to calibrate our choice model and derive four managerial insights. First, the revenue–fairness trade-off is highly nonlinear, meaning meaningful reductions in disparity can be achieved at relatively modest revenue cost, whereas approaching equal availability becomes increasingly expensive and relies more heavily on leveling the experiences of better-served size groups. Second, fairer displays do not require wholesale changes to the commercial assortment; the retailer selectively prunes options from the revenue-maximizing assortment and rotates among a modest number of complementary assortments. Third, complementary operational levers act through distinct mechanisms: size filtering can prevent unequal exposure by using customers’ size information before they choose an option, and revisit opportunities mainly recover customers after an unavailable-size encounter. Finally, although the qualitative revenue–fairness trade-off is robust

across different market environments, the effect of a given fairness weight depends on category characteristics. We therefore identify the retailer-specific revenue–fairness frontier, rather than a universal value of λ , as the relevant object for managerial decision making.

The rest of the paper is organized as follows. We review the relevant literature in Section 2. We develop the choice model, the fairness metric, and the objective function in Section 3. The structure of the optimal solution and the solution method are discussed in Section 4. Section 5 extends the baseline model to more general choice behaviors. Section 6 presents the managerial insights. Finally, we discuss possible extensions and conclude the paper in Section 7.

2. Literature Review

Our work relates to the literature on assortment optimization and fair operations. In this section, we review the relevant papers from each area of literature.

2.1. Choice Models and Assortment Optimization

The Multinomial Logit (MNL) model, developed by McFadden (1972), is widely used to model choice behavior due to its tractability. The model, however, has important limitations such as ignoring customer heterogeneity (Aouad et al. 2019) and exhibiting the ‘Independence of Irrelevant Alternatives’ (IIA) property (Blanchet et al. 2016). Among alternative models that aim to address these issues are the Mixed-Multinomial Logit (MMNL) model (McFadden and Train 2000), where customer valuations can be heterogeneous, and the consider-then-choose model (Aouad et al. 2021), where customers consider a subset of products before ranking them. We refer to Feng et al. (2022) for a comprehensive literature review. We use the MNL choice model to conceptualize how customers choose among options, while adding a second layer that captures their choices among the sizes available within each option. In this respect, our work is also related to sequential search in discrete choice models, recently studied by Kosilova and Alptekinoglu (2025).

Talluri and Van Ryzin (2004) prove that, under an MNL choice model, the optimal assortment has a revenue-ordered form. However, finding the optimal assortment across multiple customer groups using the MMNL model is NP-complete (Rusmevichientong et al. 2014). Some studies attempt to tackle this issue by finding quick, efficient heuristics (Feldman and Topaloglu 2015) or providing polynomial-time algorithms with an approximation guarantee (Désir et al. 2022). We study a case where customers in different size groups have the same valuation for an option, so we do not have this complexity issue.

Inventory stockouts are among the key factors shaping customer choice behavior. Recent research has increasingly examined how retailers can mitigate the adverse effects of stockouts. For instance, Gopalakrishnan et al. (2025) investigate the extent to which cross-store fulfillment can recover sales lost due to stockouts. At the same time, understanding how stockouts affect customer behavior

remains important. When a preferred product is unavailable, customers may substitute another product. This ‘stockout-based substitution’ behavior (Kök et al. 2015) has been studied in both the assortment optimization literature (Honhon et al. 2010) and the demand estimation literature (Musalem et al. 2010). In fashion retailing, however, stockouts are more nuanced because they can occur at both the option and SKU levels. A stockout at the SKU level results in a broken assortment, where some sizes of an option are out of stock (Caro and Gallien 2010). Estimates in Boada-Collado and Martínez-de Albéniz (2020) show that a broken assortment has a strong negative impact on sales. Li et al. (2023) study how customers react to broken assortments by analyzing a granular dataset from a shoe store chain in China. They estimate that size stockouts result in size substitution (that is, substituting with neighboring sizes) in almost 50% of the cases while only 20% of the cases lead to option substitution. Our research aims to integrate the size substitution behavior into the customer choice model and characterize the optimal assortment for a fashion retailer.

Closely related to our work is Akchen and Caro (2026), who develop a choice model that incorporates size substitution, estimate the size-substitution coefficient using transaction-level data, and solve a joint assortment and inventory optimization problem. Our work differs in three main respects. First, whereas they assume that customers assign lower valuations to broken assortments, we model broken assortments through a lower perceived fit probability. This distinction allows us to obtain closed-form solutions both with and without broken assortments. Second, they study a dynamic setting in which assortment and inventory decisions are optimized jointly, whereas we consider a static setting in which inventory availability is taken as given. Finally, we explicitly incorporate size inclusivity and fairness considerations, with the goal of preventing different size groups from experiencing systematically different levels of size unavailability.

Methodologically, we employ and adapt the *selection-based* approach. The problem of choosing the appropriate assortment is inherently combinatorial. To address this challenge, Gallego et al. (2015) develop an approach that shifts attention from the assortment displayed to the *sales quantity* of each item. They then use a duality-based algorithm to translate the resulting solution back into the original assortment-based format. In parallel, Topaloglu (2013) uses a similar idea for joint assortment and inventory decisions. At its core, this approach is a decomposition method: it substantially simplifies the problem and is particularly well suited to stochastic revenue management settings. However, because the resulting solution may involve multiple assortments, such as displaying one assortment during part of the season and another assortment later, it does not directly fit the classical static assortment optimization setting with a single assortment on display. More recently, as the idea of *randomization* has gained traction in assortment optimization, this approach has become increasingly relevant for static settings as well; see, for instance, Wang et al.

(2024) and Zhu et al. (2025). Closest to our setup, Zhu et al. (2025) study assortment optimization with market-share constraints, where the market share of a given product class must be either zero or lie within a prescribed range. After establishing the complexity of their problems, they develop approximation schemes for both randomized and deterministic policies. Our work differs from theirs in the choice model we use, the pairwise fairness structure we impose, and the extensions for which we obtain tractable optimal solutions.

2.2. Fair Operations

Two key studies by Bertsimas et al. (2011, 2012) propose ‘the price of fairness’ as a relative measure to capture the trade-off between fairness and efficiency in a resource allocation problem. Such trade-offs are studied in a variety of contexts, varying from air traffic flow management (Barnhart et al. 2012), to kidney exchange (Bertsimas et al. 2013), to portfolio management (Iancu and Trichakis 2014), to gender inequality in restroom access (Farajollahzadeh and Hu 2025). To review the literature on fair and equitable practices in operations, we refer to Karsu and Morton (2015).

An approach to incorporate fairness considerations into a decision-maker’s problem is to set a weighted average of the firm’s revenue and a measure of social responsibility as the decision-maker’s objective function. For instance, Wu et al. (2020) consider a combination of revenue and investments in socially responsible activities as the objective function. Hu et al. (2021) integrate customer surplus into a newsvendor’s profit function. Sumida et al. (2021) add customers’ utility to the objective function in an assortment optimization problem. Recent work proposes unified frameworks for convex fairness measures that enable systematic incorporation into optimization problems (Tsang and Shehadeh 2025). The objective function in our paper is a weighted average of the retailer’s revenue and a pairwise fairness metric comparing the shopping experience of different size groups.

Recent work has shown interest in incorporating fairness into assortment optimization. Levi et al. (2024) propose a rigorous definition of inclusivity in the product line design and assortment optimization. Chen et al. (2022) and Lu et al. (2023) solve a static assortment optimization problem with fairness constraints on the outcome for different products. Barré et al. (2023) follow the same goal but in a dynamic setting. While the pairwise fairness we use is inspired by Chen et al. (2022), and our ‘randomization’ approach is similar to that in Lu et al. (2023), we focus on fairness across different customer groups in this research.

3. Problem Definition

In this section, we first define the fundamentals in our model. We then outline the choice model and the firm’s assortment optimization problem subject to fairness considerations.

Retailers usually categorize similar products together, e.g., women’s T-shirts. In each category, we assume the option-SKU hierarchy we introduced in Section 1. The sale of an SKU from option j brings revenue r_j to the retailer. We index options in descending revenue order: $r_1 \geq r_2 \geq \dots \geq r_n$. We denote the *sizing system* with G , which represents the possible SKU sizes, and define $g := |G|$.

We now elaborate on the notion of *configuration*. Consider an option j that the retailer offers using a sizing system $G = \{XS, S, M, ML, L, XL, XXL\}$. However, not all sizes of an option are available all the time. So the configuration of option j , $C_j \subseteq G$, denotes the sizes in which option j is available in the inventory. For instance, if $C_j = \{S, M, XXL\}$, then option j is available in small, medium, and double extra-large sizes. Here, C_j does not represent how many of each SKU is available in the inventory; rather, $M \in C_j$ means at least one SKU of option j with size medium is available in the inventory. If there is an option $j \in N := \{1, 2, \dots, n\}$ for which we have $C_j \neq G$, we say option j has a ‘broken assortment’.

An assortment $S \subseteq N$ is the set of options that the retailer displays on its website. When the retailer includes option j in the assortment, all SKUs in C_j are displayed to the customer.¹ Online retailers, including our partner retailer, typically display all available inventory options across multiple pages on their websites. Nonetheless, when searching, customers rarely go beyond the first few pages. For instance, 73% of customers never visit the second page of a category (Avatria 2021). So, it is critical to determine which options to display on the first page. In that case, S represents the assortment of options on the first page. We summarize the core notions in our model in Definition 1.

DEFINITION 1. A configuration for option j , i.e., C_j , is the set of available sizes of that option in the inventory. An assortment S represents the set of distinct options displayed by the retailer. Finally, the broken assortment issue exists when $C_j \neq G$ for at least one $j \in N$.

3.1. The Choice Model

We develop a two-stage choice model in which customers first choose an option, then select the size to purchase. In Section 5.1, we consider a choice model in which customers first filter by size and then select among the remaining options. We model the customer journey as a sequential decision process in which customers first choose an option and subsequently evaluate size availability. This assumption aligns with the literature on customer choice with screening rules; see, e.g., Gilbride and Allenby (2004).

In a typical online apparel shopping experience, the customer first chooses the apparel category and lands on the category page shown in Figure 1a, where the first stage begins. The category page of online apparel retailers is often ‘size-agnostic,’ presenting all available options regardless

¹ We relax this assumption in Appendix EC.5.

of their configuration (we relax this assumption in Section 5.1 and assume customers first choose their sizes). We assume that the customer observes the option assortment S and, following an MNL choice model, chooses the option with the highest utility. In an MNL model, a customer's utility for option j is $U_{j,\text{customer}} = u_j + \varepsilon_{j,\text{customer}}$, where $\varepsilon_{j,\text{customer}}$ terms are i.i.d. samples of the standard Gumbel distribution. There is also a no-purchase option with u_0 . Assuming a customer chooses the option that maximizes her utility, the probability of choosing option j is:

$$\Pr(j|S) = \frac{v_j}{v_0 + \sum_{m \in S} v_m}, \quad (1)$$

where $v_j = \exp(u_j)$ is the preference weight for option j . We assume v_j is independent of the customer size, since we do not believe physical size affects customers' preferences across options. A similar assumption is applied by Li et al. (2023) and Akchen and Caro (2026).

After choosing option j , the customer lands on option j 's webpage (see Figure 1b for an example), and the second stage begins. The customers' decisions may diverge at this stage depending on their size. Each customer has an ideal size that fits her best; we call this the customer's primary size. Accordingly, we define 'size group i ' as the group of customers whose primary size is i , and denote the fraction of customers in size group i by ξ^i , such that $\sum_{i \in G} \xi^i = 1$.

Consider a customer in size group i . Upon landing on option j 's webpage, the customer evaluates C_j . If C_j contains size i , the customer purchases that SKU. However, if the primary size is unavailable, the customer may substitute with one of the SKUs in the neighboring sizes (conditional on their presence in C_j) or may leave without a purchase. Conditional on the primary size being unavailable, p^u and p^l denote the probabilities that the customer is willing to substitute one size upward and downward, respectively. In general, $p^u \neq p^l$. In practice, size substitution is localized to adjacent sizes; customers do not substitute more than one size away. In the following lemma, we define $f^i(C_j)$, the *configuration reaction coefficient*, to represent the probability that a customer in size group i purchases an SKU after observing option j 's webpage.

LEMMA 1. *The resulting purchase probability can therefore be summarized by the configuration reaction coefficient $f^i(C_j)$:*

$$f^i(C_j) := \begin{cases} 1, & \text{if } i \in C_j; \\ p^u, & \text{if } i, i-1 \notin C_j \text{ but } i+1 \in C_j; \\ p^l, & \text{if } i, i+1 \notin C_j \text{ but } i-1 \in C_j; \\ p^u + p^l, & \text{if } i \notin C_j \text{ but } \{i-1, i+1\} \subseteq C_j; \\ 0, & \text{o.w.;} \end{cases} \quad (2)$$

where $p^u, p^l \geq 0$ and $p^u + p^l \leq 1$.

Appendix EC.1 derives (2) in detail. Also, in Section 5.2, we consider the possibility of 'option substitution,' whereby customers whose desired size is unavailable may continue searching for other options rather than leaving the store immediately.

3.2. The Revenue Function and Unfairness Metric

With the choice model developed in Section 3.1, defining the retailer's revenue term is straightforward. Knowing that a fraction $f^i(C_j)$ of size group i customers buy an SKU from option j after visiting its webpage, the retailer's expected revenue from displaying assortment S to a size group i customer is:

$$\text{Rev}(S|i) = \frac{\sum_{j \in S} v_j f^i(C_j) r_j}{v_0 + \sum_{m \in S} v_m}. \quad (3)$$

Because in the baseline model the retailer cannot tailor the category webpage to the visiting customer's size, and size group i customers constitute fraction ξ^i of customers, the expected revenue from displaying assortment S is:

$$\text{Rev}(S) = \sum_{i \in G} \xi^i \frac{\sum_{j \in S} v_j f^i(C_j) r_j}{v_0 + \sum_{m \in S} v_m} = \frac{\sum_{j \in S} v_j \tilde{r}_j(C_j)}{v_0 + \sum_{m \in S} v_m}, \quad (4)$$

where $\tilde{r}_j(C_j) := r_j \sum_{i \in G} \xi^i f^i(C_j)$.

We now introduce a metric to measure unfairness in customer experience. To do so, we need an index of the experiences of different customer size groups regarding size availability. In Section 3.1, we noted that a fraction $(1 - f^i(C_j))$ of size group i customers who visit option j 's webpage do not purchase an SKU because of size unavailability. Therefore, we define unavailability, $U^i(S)$, as the probability that a size- i customer leaves without a purchase:

$$U^i(S) := \frac{v_0}{v_0 + \sum_{m \in S} v_m} + \frac{\sum_{j \in S} v_j (1 - f^i(C_j))}{v_0 + \sum_{m \in S} v_m}. \quad (5)$$

From the retailer's perspective, $U^i(S)$ is a practical measure because the probability that a customer leaves the store empty-handed can be estimated from clickstream data. Moreover, customers often evaluate assortment fairness based on whether they (compared to others) can find and purchase a suitable option, which $U^i(S)$ captures. Finally, the methods we develop under $U^i(S)$ can be readily adapted to any performance metric of the form $(\sum_{j \in S} \phi_j^i v_j) / (v_0 + \sum_{j \in S} v_j)$, where ϕ_j^i is an arbitrary value. For example, the measure $(\sum_{j \in S} \mathbf{1}(i \in C_j) v_j) / (v_0 + \sum_{j \in S} v_j)$ can be used when customers are likely to purchase an option only if it is available in their primary size and do not substitute with neighboring sizes. Note that in the baseline model, the outside option component is common across size groups, so pairwise differences in $U^i(S)$ isolate disparities arising from size availability.

Unfairness in a firm's operations can be measured in various ways. One approach is to consider the gap between the payoffs received by the best-off and worst-off customers. This is the *pairwise fairness* measure applied in designing recommendation algorithms in Beutel et al. (2019) and Narasimhan et al. (2020). Most relevant to our work, Chen et al. (2022) apply this measure in an assortment optimization setting, where an online platform ensures that its different sellers

receive a fair share of visibility on the retailer’s website. The popularity of pairwise fairness stems from its straightforward interpretation and ease of integration into mathematical models (Karsu and Morton 2015). Following this line of logic, we use the gap between the highest and lowest unavailability among size groups as the unfairness metric for an assortment:

$$\text{Unf}(S) := \max_{l,k \in G} [U^k(S) - U^l(S)]. \quad (6)$$

An alternative measure is the *Rawlsian* metric, whereby the maximum unavailability across size groups represents the level of unfairness in a system (Farajollahzadeh and Hu 2025). One can easily extend our results to apply the Rawlsian metric by introducing a ‘fictitious’ customer group f without any impact on the revenue (i.e., $\xi^f = 0$) and $U^f(S) = 0$ for any assortment S .

3.3. The Objective Function

There are two common approaches to incorporating behavioral considerations into an optimization problem. In one approach, the anticipated behavioral consequences of a pricing or assortment decision are embedded in customer utility and the aggregate demand function (Popescu and Wu 2007, Wang 2018, Hu and Nasiry 2018). In a second approach, the firm (or a central planner) imposes constraints on pricing and assortment decisions to satisfy regulatory requirements and/or prevent (or limit) customer backlash and reputational damage from practices perceived as unfair or unequal (Sumida et al. 2021, Cohen et al. 2022, Li et al. 2024), with the goal of ensuring inclusivity. Our notion of fairness aligns with the second approach and is closely related to inclusivity in retail practices because it aims to reduce the systematic exclusion of certain size groups from being served by the assortment. At the same time, it is conceptually distinct from several related ideas. First, our metric does not model customer satisfaction or disappointment directly, which would require a utility-based choice framework and the estimation of additional behavioral parameters. Second, it differs from accessibility: in our model, all customers have access to the assortment, and the issue we address is unequal service outcomes conditional on access. Finally, our fairness concept differs from equity. While equity emphasizes differential treatment based on heterogeneity across size groups in the customer base, our baseline metric penalizes disparities symmetrically across size groups.

Unfair practices in assortment decisions have two possible effects on the retailer. First, social media makes it easier for customers to become aware of non-inclusive and unfair practices. Second, such perceived unfairness may influence customers’ future purchase decisions. As Bateni et al. (2022) put it, fairness is a “crucial system objective” in settings where a profit-maximizing platform seeks to ensure repeat purchases by treating customers fairly. Therefore, given the competitive

fashion retail landscape where customer retention is a strategic concern, we define the retailer's *fairness-adjusted revenue* as follows:

$$\Pi(S) := \text{Rev}(S) - \lambda \text{Unf}(S), \quad (7)$$

where $\lambda \geq 0$ represents the relative importance of unfairness concerns compared to revenue. As λ increases, the retailer places more emphasis on fairness in selecting its assortment. Customers form their loyalty decision during the first 60 seconds of interacting with the website (*Forbes* 2016), and a negative impression created by size unavailability can therefore lead to attrition. As such, λ can be interpreted as the retailer's marginal willingness to trade revenue for reduced unfairness and thus less customer attrition.²

We write the retailer's deterministic assortment optimization problem as:

$$\Pi_D = \max_{S \subseteq [n]} \Pi(S) \quad (\text{AOP-DetFair})$$

where (AOP-DetFair) stands for *Assortment Optimization Problem - Deterministically Fair*, and Π_D denotes the optimal fairness-adjusted revenue of a deterministic approach.

Whereas (AOP-DetFair) aligns with the classic assortment optimization problems in a physical setting, in online retailing, which is our focus, it takes an additional layer. Unlike a physical retailer, an online retailer is not limited to offering only one assortment to all arriving customers, and can *randomize* over a set of assortments, meaning it can choose a probability distribution over the feasible assortments and display an assortment sampled from this distribution. With the rise of complex optimization methods in online retailing, randomization in assortment optimization has gained traction in recent years (see e.g., Chen et al. (2022), Wang et al. (2024), Zhu et al. (2025)). For instance, Wang et al. (2024) apply it to robustify the retailer's revenue against estimation errors. More recently, Zhu et al. (2025) employ randomization to solve an assortment optimization with different suppliers where market share constraints must be satisfied.

With randomization, the retailer decides $\mathbf{P}$, a vector of length 2^n representing the display probability of each feasible assortment, meaning that $\mathbf{P} \geq \mathbf{0}$ and $\mathbf{P}^\top \mathbf{1} = 1$. We denote the probability of displaying assortment S with $P(S)$. With randomization, we define (in italics) $\text{Rev}(\mathbf{P}) := \sum_{w=1}^{2^n} P(S^w) \text{Rev}(S^w)$ and $U^i(\mathbf{P}) := \sum_{w=1}^{2^n} P(S^w) U^i(S^w)$ to display the *expected revenue*

² An alternative interpretation of λ is as the Lagrange multiplier of a hard-constraint formulation in which the retailer maximizes expected revenue subject to $\max_{l,k \in G} [U^k(S) - U^l(S)] \leq F_{\max}$, where $F_{\max}$ is the largest tolerable gap in unavailability across size groups. By varying $F_{\max}$ and computing the associated multiplier λ^F , the retailer can trace a correspondence between its fairness tolerance and the implied value of λ . This correspondence offers a practical calibration strategy: to enforce a desired fairness tolerance (e.g., $F_{\max} = 0.10$), the retailer can solve the hard-constraint problem at that tolerance and use the resulting λ^F in the soft-penalty formulation (7).

and *expected unavailability*, respectively. Therefore, the retailer's fairness-adjusted revenue under randomization is:

$$\Pi(\mathbf{P}) := \text{Rev}(\mathbf{P}) - \lambda \text{Unf}(\mathbf{P}), \quad (8)$$

where $\text{Unf}(\mathbf{P}) := \max_{l,k \in G} [U^k(\mathbf{P}) - U^l(\mathbf{P})]$. Similarly, we define the retailer's assortment optimization problem under randomization as:

$$\Pi_R = \max_{\mathbf{P} \geq \mathbf{0}, \mathbf{P}^\top \mathbf{1} = 1} \Pi(\mathbf{P}). \quad (\text{AOP-RandFair})$$

The name of (AOP-RandFair) stands for Assortment Optimization Problem - Randomized Fair, and Π_R denotes the optimal fairness-adjusted revenue of a randomized approach. We now prove a structural property that (AOP-DetFair) and (AOP-RandFair) have in common.

LEMMA 2. In (AOP-DetFair) and (AOP-RandFair): (i) The unfairness term is non-negative. (ii) Fairness-adjusted revenue, revenue, and unfairness decrease in λ in the optimal solution.

Proof: Proofs are in Appendix EC.2.

The first part of Lemma 2 is intuitive given the notion of pairwise fairness in our model. As λ becomes larger, more focus is allocated to decreasing the unfairness term, thereby decreasing both revenue and unfairness. Obviously, then, fairness-adjusted revenue decreases in λ .

Randomization allows (AOP-RandFair) to attain higher fairness-adjusted revenue than (AOP-DetFair). Example 1 below illustrates the value of randomization in the context of our problem.

EXAMPLE 1. Suppose there are two customer groups l and k and two feasible assortments S^1 and S^2 with $\text{Rev}(S^1) = 10$ and $\text{Rev}(S^2) = 8$. Also assume that $U^l(S^1) = 0.75$, $U^k(S^1) = 0.25$, $U^l(S^2) = 0.25$, and $U^k(S^2) = 0.75$. Note that $U^k(S^1) - U^l(S^1) = -0.5$ and $U^k(S^2) - U^l(S^2) = 0.5$. Define $\pi = P(S^1)$ and observe that the expected fairness-adjusted revenue is:

$$\Pi_R(\pi) = 10\pi + 8(1 - \pi) - \lambda \max\{0.5\pi - 0.5(1 - \pi), -0.5\pi + 0.5(1 - \pi)\}. \quad (9)$$

We solve (9) for two values of λ . If $\lambda = 1$, the retailer's optimal decision is $\pi^*(\lambda = 1) = 1$, which is equivalent to solely displaying S^1 . Therefore, the solutions to (AOP-DetFair) and (AOP-RandFair) with $\lambda = 1$ are the same, with $\Pi_R^* = \Pi_D^* = 9.5$. Nonetheless, with $\lambda = 3$, the optimal decision is $\pi^*(\lambda = 3) = 0.5$. In this situation, we have $\Pi_R^* = 9$ while $\Pi_D^* = 8.5$, representing the advantage of randomization.

To better see this, note that if only S^1 or only S^2 is shown, the impact of the unfairness term in (9) is equal to -0.5λ . Therefore, solely displaying S^2 is never optimal for the retailer, as assortment S^1 has the same level of unfairness as S^2 but yields more revenue. For $\lambda = 1$, setting $\pi < 1$ and

displaying S^2 with a non-zero probability causes a decrease in the revenue that has a stronger effect on the fairness-adjusted revenue than alleviating the unfairness. Nevertheless, for $\lambda = 3$, the impact of the unfairness term on the fairness-adjusted revenue is more pronounced, and the retailer prefers to incur a drop in its revenue through randomization. ■

In Example 1, the optimal solution to (AOP-DetFair) creates a *systematic disparity* between the two customer groups. However, with randomization and both assortments displayed with non-zero probabilities (here, 50%), the unfairness in one assortment is canceled out by the advantage in the other. With randomization, the systematic disparity between the two customer groups can be alleviated in expectation. Let us define fairness as the complement of the unfairness term (i.e., $Fair(\mathbf{P}) := 1 - Unf(\mathbf{P})$). We formally discuss the impact of randomization in the following lemma.

LEMMA 3. *With randomization: (i) The unfairness term is convex in $\mathbf{P}$ while the revenue term is linear in it. (ii) The revenue-fairness frontier is concave with respect to the origin.*

The first part of Lemma 3 shows that randomizing across multiple assortments can reduce expected unfairness, while expected revenue remains the weighted average of the revenues of the underlying assortments. Combined with Lemma 2, this implies a concave revenue–fairness frontier, as established in the second part of Lemma 3. We provide more insights in Section 6.

4. Characterization of the Optimal Solution

In this section, we characterize the structure of the optimal solution to (AOP-RandFair). We first study the special case without a broken assortment. Then, we move to the general case with a broken assortment and explain how our method imposes fairness using a stylized example. Afterward, we develop the solution method for the general case.

4.1. Optimal Solution without a Broken Assortment

Without a broken assortment, the retailer can present every SKU of every option it so desires. A classic result in the AOP literature (Talluri and Van Ryzin 2004) is that the option assortment that maximizes $(\sum_{j \in S} r_j v_j) / (v_0 + \sum_{j \in S} v_j)$ follows a nested revenue-ordered structure. Suppose this problem’s optimal assortment is the set $S^{nb} = \{1, \dots, m\}$.

We establish the following result on the optimal assortment in the absence of a broken assortment.

LEMMA 4. *Without a broken assortment, S^{nb} is an optimal solution for (AOP-RandFair).*

Because S^{nb} is a single assortment (without any randomization), a direct result of Lemma 4 is that S^{nb} is also optimal for (AOP-DetFair).

Lemma 4 identifies the operational source of the revenue–fairness tension in our setting. When every option has a complete size configuration, there is no difference between size groups’ availability experience, irrespective of the displayed assortment. The unfairness term therefore vanishes,

and the revenue-maximizing solution and the fairness-adjusted revenue-maximizing solution fully align regardless of λ . Hence, fairness considerations affect the assortment decision only when the retailer inherits broken size configurations. In this sense, broken size runs create the need for the fairness-aware display intervention studied in this paper.

4.2. The Fairness Mechanism with a Broken Assortment

Size unavailability and broken assortments are endemic to fashion retailing (*Retail-Week* 2022). With a broken assortment, displaying an option with size stockouts affects size groups differently, distorting the alignment between revenue and fairness. The assortment decision therefore affects both revenue and the unfairness term, and the tradeoff depends on the value of λ . Before developing the optimization method, we illustrate how a fairness-aware policy uses this display lever using an example.

EXAMPLE 2. In Figure 2, we display an inventory of eight options. Each column corresponds to an option, and each row corresponds to a size. An entry of 1 indicates that a size–option pair is offered; 0 indicates that the size is unavailable for that option. For instance, Option 1 is available in all sizes, whereas Option 4 is available only in sizes M and L. In this example, to isolate the impact of size coverage, we set the revenues (r_j ’s) and preference weights (v_j ’s) of all options equal. However, we include a representative range of configurations: from available in all sizes (Option 1), to available only in major sizes (Option 3), to available only in marginal sizes (Option 6), etc. We also assume a symmetric triangular customer size distribution, so middle sizes (M and L) have the largest populations and marginal sizes (XS and XXL) have the smallest.³

	O ₁	O ₂	O ₃	O ₄	O ₅	O ₆	O ₇	O ₈
XS	1	0	0	0	1	1	1	0
S	1	1	0	0	1	1	1	0
M	1	1	1	1	1	0	0	0
L	1	1	1	1	1	0	0	0
XL	1	1	0	1	0	1	0	1
XXL	1	0	0	1	0	1	0	1

Figure 2 Inventory availability of options and sizes.

We then report snapshots of optimal assortments (assortments that maximize fairness-adjusted revenue) at three representative values of λ : low ($\lambda = 0$), medium ($\lambda = 0.25$), and high ($\lambda = 0.5$) in Figure 3. The selected options are distinguished with the color green. We also report the unavailability (U^i) for each size group under the optimal assortment in the last column. To avoid the complexity of displaying randomization, we designed this example so the optimal solution requires no randomization.

³ More specifically, we set $r_j = 1, v_j = 1$ for all options, $v_0 = 0.649$, $p^u = p^l = 0.25$, and $\xi = \{0.05, 0.2, 0.25, 0.25, 0.2, 0.05\}$.

	O ₁	O ₂	O ₃	O ₄	O ₅	O ₆	O ₇	O ₈	U^i
XS	1	0	0	0	1	1	1	0	0.52
S	1	1	0	0	1	1	1	0	0.30
M	1	1	1	1	1	0	0	0	0.14
L	1	1	1	1	1	0	0	0	0.14
XL	1	1	0	1	0	1	0	1	0.30
XXL	1	0	0	1	0	1	0	1	0.52

(a)

	O ₁	O ₂	O ₃	O ₄	O ₅	O ₆	O ₇	O ₈	U^i
XS	1	0	0	0	1	1	1	0	0.42
S	1	1	0	0	1	1	1	0	0.25
M	1	1	1	1	1	0	0	0	0.25
L	1	1	1	1	1	0	0	0	0.25
XL	1	1	0	1	0	1	0	1	0.25
XXL	1	0	0	1	0	1	0	1	0.42

(b)

	O ₁	O ₂	O ₃	O ₄	O ₅	O ₆	O ₇	O ₈	U^i
XS	1	0	0	0	1	1	1	0	0.35
S	1	1	0	0	1	1	1	0	0.30
M	1	1	1	1	1	0	0	0	0.30
L	1	1	1	1	1	0	0	0	0.30
XL	1	1	0	1	0	1	0	1	0.30
XXL	1	0	0	1	0	1	0	1	0.35

(c)

Figure 3 Optimal assortments for $\lambda = 0$ (a), $\lambda = 0.25$ (b), and $\lambda = 0.5$ (c).

At low λ , unavailability is disproportionately higher for marginal sizes, implying a large unfairness. By gradually increasing λ , the retailer reduces disparity through two distinct channels. First, the policy can *lift* the experience of initially disadvantaged size groups by displaying options that better cover those groups. Second, it can *level* experiences by suppressing options that disproportionately benefit initially advantaged groups. Both channels are present when we increase λ from 0 to 0.25: By adding Option 6 (available in all sizes except M and L), unavailability decreases for the marginal sizes (lifting channel) while increasing for the initially best-served middle sizes (leveling channel). Similarly, by increasing λ to 0.5, the retailer further decreases unfairness by dropping Option 2 (available in all sizes except XS and XXL), despite Option 2's relatively strong contribution to expected revenue.

As this example illustrates, fairness improves not because every size is offered for every option, but because the retailer combines options with complementary size configurations, thereby balancing customers' exposure to size unavailability. For instance, while Option 4 does not contain S and XS sizes, Option 5 omits XL and XXL sizes, thereby balancing representation of sizes across the final assortment. ■

4.3. Optimal Solution with a Broken Assortment

We now solve (AOP-RandFair) under a broken assortment. We introduce a framework called the *selection-based* method, which comprises two parts: a *forward transformation*, which converts the original combinatorial assortment problem into an LP in terms of selection probabilities, and a *backward transformation*, which maps the LP solution back to an optimal randomized assortment policy.

We begin the forward transformation phase by linearizing the unfairness metric using an auxiliary variable ψ that upper-bounds all pairwise fairness differences. This yields the following equivalent

formulation:

$$\begin{aligned}
\Pi &= \max_{\mathbf{P}, \psi} \sum_{S \subseteq N} P(S) \frac{\sum_{j \in S} v_j \tilde{r}_j(C_j)}{v_0 + \sum_{m \in S} v_m} - \psi \\
\text{s.t. } \quad \psi &\geq \lambda \sum_{S \subseteq N} P(S) \left[\frac{v_0 + \sum_{j \in S} v_j (1 - f^k(C_j))}{v_0 + \sum_{m \in S} v_m} - \frac{v_0 + \sum_{j \in S} v_j (1 - f^l(C_j))}{v_0 + \sum_{m \in S} v_m} \right] \quad \forall k, l \in G, \\
\sum_{S \subseteq N} P(S) &= 1, \quad P(S) \geq 0 \quad \forall S \subseteq N,
\end{aligned}$$

(AOP-Assortment)

where (AOP-Assortment) stands for Assortment Optimization Problem - Assortment-Based. It is straightforward to show that without fairness considerations (i.e., $\lambda = 0$), the optimal solution is a singleton with a $\tilde{r}_j$ -ordered structure. However, in the broken-assortment case, option revenues must be adjusted to account for size availability. Therefore, the main complexity is incorporating fairness considerations.

Although this step removes the max operator from the objective, the formulation remains computationally challenging for two reasons: (i) the decision variable $P(S)$ is defined over exponentially many assortments, and (ii) the revenue term is fractional in these variables. To address both challenges simultaneously, we replace the assortment-based decision variables (i.e., the probabilities of showing individual assortments) with option-selection probabilities. Specifically, for each option $j \in N$, we define x_j as the probability that a customer selects option j , and let x_0 denote the probability of selecting the outside option. The total-probability constraint in the new formulation is $\sum_{j \in N} x_j + x_0 = 1$. Under this reformulation, expected revenue is $\sum_{j \in N} \tilde{r}_j(C_j) x_j$, and the fairness constraints become:

$$\psi \geq \lambda \sum_{j \in N} x_j (f^k(C_j) - f^l(C_j)) \quad \forall k, l \in G.$$

Finally, to ensure that selection probabilities align with the structure imposed by an MNL choice model, we want to set x_j so that $x_j = \sum_{S \subseteq N} P(S) \Pr(j|S)$, where $\Pr(j|S)$ follows the structure in (1). For any $j \in N$ and assortment S , the probability of choosing option j is $\Pr(j|S) = v_j / (v_0 + \sum_{m \in S} v_m)$ if $j \in S$, and zero otherwise. In contrast, regardless of the assortment, the probability of choosing the outside option is $\Pr(0|S) = v_0 / (v_0 + \sum_{m \in S} v_m)$. Therefore, if the optimal solution consists of displaying a single assortment S (i.e., $P(S) = 1$), then the following relation between x_j and x_0 holds:

$$x_j = \Pr(j | S) = \begin{cases} \frac{v_j}{v_0} \Pr(0 | S) = \frac{v_j}{v_0} x_0, & \text{if } j \in S; \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

With randomization, the gap between x_j and $(v_j/v_0)x_0$ can only increase, as option j may be absent from some assortments, while the outside option is part of all assortments. As such, the correct scaling between purchase and no-purchase probabilities is enforced by

$$\frac{x_j}{v_j} \leq \frac{x_0}{v_0}, \quad \forall j \in N. \quad (11)$$

Combining these components yields the following LP:

$$\begin{aligned} \Pi^{\text{Selection-Based}} &:= \max_{\mathbf{x}, \psi} \sum_{j \in N} \tilde{r}_j(C_j) x_j - \psi \\ \text{s.t.} \quad \psi &\geq \lambda \sum_{j \in N} x_j (f^k(C_j) - f^l(C_j)) \quad \forall k, l \in G, \\ \frac{x_j}{v_j} &\leq \frac{x_0}{v_0} \quad \forall j \in N, \\ \sum_{j \in N} x_j + x_0 &= 1, \\ x_j &\geq 0 \quad \forall j \in N \cup \{0\}, \end{aligned} \quad (\text{AOP-Selection})$$

where (AOP-Selection) stands for *Assortment Optimization Problem - Selection-Based*.

This LP has a polynomial number of variables and constraints and can be solved efficiently using standard solvers. Methods for linearizing and solving fractional optimization problems date back to Charnes and Cooper (1962). Nevertheless, showing that the optimal selection probabilities from (AOP-Selection) can be translated into actionable assortment-based decisions is nontrivial. To establish full equivalence with the original assortment-based formulation, we use a backward transformation method developed in Topaloglu (2013) and Gallego et al. (2015), and tailored to constrained assortment optimization problems with randomization in Zhu et al. (2025).

Let $\hat{x}_j$, $j \in N$, be the optimal solution to (AOP-Selection). An option can have a higher selection probability not because it must be displayed more, but because it has a higher preference weight. We account for this effect by dividing $\hat{x}_j$ by v_j . Re-index the options to have $\hat{x}_1/v_1 \geq \hat{x}_2/v_2 \geq \dots \geq \hat{x}_n/v_n$. By constraint (11), $\hat{x}_0/v_0 \geq \hat{x}_1/v_1 \geq \hat{x}_2/v_2 \geq \dots \geq \hat{x}_n/v_n$. Now define nested sets $S_m := \{1, \dots, m\}$, with $S_0 = \emptyset$. The critical step is to assign the correct display probability to each set. Let $\hat{\mathbf{P}}$ be a candidate solution to (AOP-Assortment). Set $\hat{P}(S) = 0$ for any assortment $S \notin \{S_0, S_1, \dots, S_n\}$. We must design $\hat{\mathbf{P}}$ so that

$$\sum_{m=0}^n \hat{P}(S_m) \Pr(j \mid S_m) = \hat{x}_j, \quad \forall j \in N \cup \{0\}. \quad (12)$$

Start with option n , because it appears only in S_n . By equation (12), $\hat{P}(S_n)$ is uniquely defined:

$$\hat{P}(S_n) = \frac{\hat{x}_n}{v_n} \left[v_0 + \sum_{m \in S_n} v_m \right].$$

Move to option $n - 1$ which appears in two assortments S_{n-1} and S_n . By equation (12):

$$\hat{P}(S_{n-1}) \frac{v_{n-1}}{v_0 + \sum_{m \in S_{n-1}} v_m} + \hat{P}(S_n) \frac{v_{n-1}}{v_0 + \sum_{m \in S_n} v_m} = \hat{x}_{n-1}$$

or equivalently:

$$\hat{P}(S_{n-1}) = \left(\frac{\hat{x}_{n-1}}{v_{n-1}} - \frac{\hat{x}_n}{v_n} \right) \left[v_0 + \sum_{m \in S_{n-1}} v_m \right].$$

We follow the same approach for all options and obtain:

$$\hat{P}(S_j) = \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) \left[v_0 + \sum_{i \in S_j} v_i \right] \quad \forall j \in N \cup \{0\}, \quad (13)$$

where we define $\hat{x}_{n+1} = 0$ for notational convenience.

By showing that the optimal solutions of (AOP-Assortment) and (AOP-Selection) are feasible for each other, and using the transformed probabilities $\hat{\mathbf{P}}$, we prove the following theorem.

THEOREM 1. (i) *The optimal values of (AOP-Assortment) and (AOP-Selection) are equal.* (ii) *Allocation $\hat{\mathbf{P}}$ attains the optimal solution of (AOP-Assortment).* (iii) *Expected revenue and unfairness are identical in the two formulations.*

Part (iii) of Theorem 1 sheds light on the analytical value of the selection-based methodology. If the retailer is only interested in the value of the revenue or the unfairness term (rather than its assortment-based implementation), for instance, if the retailer is evaluating the effect of different values of λ , it can get the optimal revenue and unfairness values without doing the backward stage.

To further assess the value of incorporating fairness considerations into the optimization model, we ran several simulations calibrated on real data, which we elaborate on in Section 6. Nonetheless, our simulations show that unfairness can be drastically reduced with only a small drop in revenue.

We now present structural results on the optimal assortment $\hat{\mathbf{P}}$ for (AOP-Assortment), building on the optimal solution to (AOP-Selection).

LEMMA 5. *There exists an optimal $\hat{\mathbf{P}}$ with at most g^2 distinct assortments.*

Regardless of optimality, if a solution suggests randomizing over a large number of assortments, it is impractical. Looking at (13), one may fear that $\hat{\mathbf{P}}$ may contain n distinct assortments. Lemma 5 addresses this concern by showing that $\hat{\mathbf{P}}$ needs at most g^2 distinct assortments. Although the nested construction generates up to n candidate assortments, whenever adjacent products share the same $\hat{x}/v$, the corresponding nested assortments represent the same treatment tier and thus collapse into a single effective assortment. Consequently, the support size is determined not by the number of products, but by the number of distinct breakpoints in $\{\hat{x}_0/v_0, \hat{x}_1/v_1, \hat{x}_2/v_2, \dots, \hat{x}_n/v_n\}$. Because an optimal basic feasible solution to Problem (AOP-Selection) needs at most g^2 such breakpoints to balance all pairwise fairness constraints, the induced randomized assortment policy requires at most g^2 distinct assortments.

5. Extensions

In this section, we extend our model and solution method in two practical directions. First, we consider the case where customers choose their sizes immediately after landing on the category page, before viewing any options. Second, we consider option-substitution behavior, whereby customers facing size stockouts may opt to visit the store again. Additionally, we developed extensions that integrate size-based filtering with option substitution and endogenize the configuration choice; for brevity, we present these materials in the appendix.

5.1. Size Filtering Possibility

Our baseline model assumes that customers first choose an option and then evaluate the sizes available for that option. However, some online retailers allow shoppers to filter by size before browsing. In this approach, customers first select a size, then choose among the available products. We call this the ‘size-filtering’ or ‘size-first’ model, interchangeably.

Before introducing the size-filtering model, it is useful to clarify how unfairness differs from the baseline. In the baseline, unfairness arises when a customer selects a product shown on the category page, only to find that it is unavailable in her size. In the size-filtering model, customers choose their size first, then browse products available in that size. Here, unfairness arises because some size groups may face much smaller choice sets than others.⁴

In the size-filtering approach, the customer chooses her size (e.g., via a size-filter combobox) after entering the category webpage. The choice among the represented options follows an MNL selection process. The customer then buys an SKU from the selected option with probability $f^i(C_j)$. By choosing her size first, and in contrast to the baseline model, the customer allows the retailer to curate the assortment based on the revealed size preference. We denote the assortment curated to customers in size group i with $\mathbf{P}^i$. The retailer’s problem is then to design and offer g distributions: $\mathcal{P} := \{\mathbf{P}^1, \dots, \mathbf{P}^g\}$. Because each size group faces an assortment curated to their size, the expected revenue from assortment distribution $\mathcal{P}$ is:

$$\text{Rev}(\mathcal{P}) = \sum_{i \in G} \xi^i E_{P^i} \text{Rev}(S | i) = \sum_{i \in G} \xi^i \sum_{S \subseteq N} P^i(S) \text{Rev}(S | i). \quad (14)$$

Similarly, the expected unfairness from assortment distribution $\mathcal{P}$ is:

$$\text{Unf}(\mathcal{P}) = \max_{l, k \in G} [E_{P^k} U^k(S) - E_{P^l} U^l(S)] = \max_{l, k \in G} \left[\sum_{S \subseteq N} P^k(S) U^k(S) - \sum_{S \subseteq N} P^l(S) U^l(S) \right]. \quad (15)$$

⁴ A customer expressed this frustration as follows: “(before filtering by size) The little button says Review 250 items, you click L & XL, button now says: Review 47 items. Like seriously?!?” (Reddit 2024).

Putting these components together, the retailer's assortment problem under size-first customer search behavior can be written as:

$$\begin{aligned}
\Pi_{\text{Size-Filtering}} &= \max_{\mathcal{P}, \psi} \sum_{i \in G} \xi^i E_{P^i} \text{Rev}(S \mid i) - \psi \\
\text{s.t. } \quad &\psi \geq \lambda [E_{P^k} U^k(S) - E_{P^l} U^l(S)] \quad \forall k, l \in G, \\
&\sum_{S \subseteq N} P^i(S) = 1 \quad \forall i \in G, \\
&P^i(S) \geq 0 \quad \forall i \in G, S \subseteq N,
\end{aligned} \tag{SFP-Assortment}$$

where (SFP-Assortment) stands for Size-Filtering Problem - Assortment-Based. Because in this problem the retailer must find the optimal assortment for each customer group, the number of decision variables is $g \times 2^n$. To solve the problem efficiently, we need to apply a transformation similar to that used in the baseline model. Specifically, we introduce choice probability vectors $\vec{X} := \{X^1, \dots, X^g\}$, where $X^i := \{x_1^i, \dots, x_j^i, \dots, x_n^i\}$ represents the probability vector that a customer in size group i chooses option j in product set N . The probability of no-purchase from a customer in size group i is denoted by x_0^i . We follow the same steps as the baseline model to develop the selection-based problem:

$$\begin{aligned}
\Pi_{\text{Size-Filtering}}^{\text{Selection-Based}} &:= \max_{\mathbf{X}, \psi} \sum_{i \in G} \xi^i \sum_{j \in N} r_j f^i(C_j) x_j^i - \psi \\
\text{s.t. } \quad &\psi \geq \lambda \left[\sum_{j \in N} f^k(C_j) x_j^k - \sum_{j \in N} f^l(C_j) x_j^l \right] \quad \forall k, l \in G, \\
&\frac{x_j^i}{v_j} \leq \frac{x_0^i}{v_0} \quad \forall j \in N, \forall i \in G, \\
&\sum_{j \in N} x_j^i + x_0^i = 1 \quad \forall i \in G, \\
&x_j^i \geq 0 \quad \forall j \in N \cup \{0\}, i \in G,
\end{aligned} \tag{SFP-Selection}$$

where (SFP-Selection) stands for Size-Filtering Problem - Selection-Based. The above LP determines the optimal selection probabilities under the retailer's fairness-adjusted revenue objective. These selection probabilities can be converted into randomized assortments using a method similar to that in (13). Let $\hat{\mathcal{P}} := \{\hat{P}^1, \dots, \hat{P}^g\}$ denote the resulting assortment policy. The following theorem establishes the validity of this transformation by showing that the resulting assortment policy obtained from the LP program in (SFP-Selection) also maximizes the original assortment problem in (SFP-Assortment) and vice versa.

THEOREM 2. (i) The optimal values of (SFP-Assortment) and (SFP-Selection) are equal. (ii) Allocation $\hat{\mathcal{P}}$ attains the optimal solution of (SFP-Assortment). (iii) Expected revenue and unfairness are identical in the two formulations.

Note that the baseline model is a more constrained case of the size-first approach. This becomes clear by adding the set of constraints $\mathbf{P}^l = \mathbf{P}^k$ for every pair of size groups to (SFP-Assortment), which means the retailer displays the same assortment to all customers regardless of their selected sizes. Therefore, the optimal objective value of the size-first approach exceeds that of the baseline, either due to higher revenue, lower unfairness, or both. In Section 6.3, we discuss the impact of this extension more extensively.

5.2. Option Substitution

In this extension, we incorporate the possibility that customers may revisit the category page after facing size unavailability for the option they first selected. That is, customers may substitute with other options in case of size stockouts. Implementing this possibility requires extending the customer choice model, the retailer's assortment decisions, and the revenue and unfairness terms. We call this model 'option substitution' or 'revisit behavior' interchangeably.

In the baseline model, a customer from size group i selects option j and evaluates configuration C_j . If C_j contains her preferred size, she purchases an SKU and exits. Otherwise, she substitutes to neighboring sizes with probability $f^i(C_j)$ or leaves the store with probability $1 - f^i(C_j)$. In the revisit model, if the chosen option does not contain her preferred size, she returns to the category page with probability δ and repeats the same process. Hence, after the first visit, she purchases from option j with probability $f^i(C_j)$, revisits with probability $\delta(1 - f^i(C_j))$, and leaves with probability $(1 - \delta)(1 - f^i(C_j))$.

In the baseline model, the retailer offers a single assortment throughout the customer's visit. With revisits, however, the retailer can show a different assortment when the customer returns to search for another option. Because the retailer observes the customer's initial choice, this follow-up assortment can depend on that information. We therefore model a two-stage assortment policy: a first-stage randomized assortment with distribution $\mathbf{P}_1$, and a second-stage randomized assortment with distribution $\mathbf{P}_{2,(j)}$, shown to customers who return after not purchasing option j in the first stage.

Under the option-substitution model, customers may observe two assortments over their search cycle. As a result, unfairness and revenue must be calculated over the entire search cycle. We therefore define two-stage unavailability for size group i with $U^i(S_1, S_2)$, which represents the level of unavailability for a customer in size group i who is faced with assortment S_1 in the first stage and S_2 in the second stage. Because a customer may encounter size unavailability more than once when revisits are allowed, $U^i(S_1, S_2)$ measures the expected number of unavailable-size encounters per entering size- i customer over the entire search cycle, rather than a terminal no-purchase probability; when $\delta = 0$, it reduces to the baseline measure. Similarly, we define $\text{Rev}^i(S_1, S_2)$ to denote the

expected revenue generated when a customer group of size i is exposed to assortments S_1 and S_2 during her search cycle. The full expressions for two-stage unavailability and revenue are provided in Appendix EC.3.

Whereas the first-stage assortment distribution is the same among all customers, the second-stage distribution depends on the first option a customer visits. Therefore, the retailer must decide $\mathcal{P} := \{\mathbf{P}_1, \mathbf{P}_{2,(1)}, \dots, \mathbf{P}_{2,(n)}\}$, which entails $(1+n) \times 2^n$ decision variables. Following the pairwise approach for estimating unfairness of assortments, the retailer's assortment optimization problem with revisits can be written as:

$$\begin{aligned} \Pi_{\text{Revisit}} = & \max_{\psi, \mathbf{P}_1, \mathbf{P}_{2,(1)}, \dots, \mathbf{P}_{2,(n)}} \sum_{i \in G} \xi^i E_{P_1} \sum_{j \in N} E_{P_{2,(j)}} \text{Rev}^i(S_1, S_2) - \psi \\ \text{s.t. } & \psi \geq \lambda \left[E_{P_1} \sum_{j \in N} E_{P_{2,(j)}} [U^k(S_1, S_2) - U^l(S_1, S_2)] \right] \quad \forall k, l \in G, \\ & \sum_{S_1 \subseteq N} P_1(S_1) = 1, \quad \sum_{S_2 \subseteq N} P_{2,(j)}(S_2) = 1 \quad \forall j \in N, \\ & P_1(S_1), P_{2,(j)}(S_2) \geq 0 \quad \forall S_1 \subseteq N, S_2 \subseteq N, j \in N, \quad P_{2,(j)}(S_2) = 0 \quad \text{if } j \in S_2. \end{aligned} \quad (\text{RBP-Assortment})$$

where (RBP-Assortment) stands for *Revisit Behavior Problem - Assortment-Based*. Note that the last constraint, $P_{2,(j)}(S_2) = 0$ if $j \in S_2$, ensures the second-stage assortment does not include the option the customer already visited in the first stage, since we know the customer would not choose that option.

To solve an optimization problem of this size efficiently, we apply the same selection-based transformation used in the baseline model. Specifically, we introduce a first-stage selection vector $X = \{x_1, \dots, x_n\}$ and, for the option-substitution model, a set of second-stage selection vectors $\vec{Y} = \{Y^1, \dots, Y^g\}$, where $Y^i = \{y_1^i, \dots, y_n^i\}$ for each $i \in G$. Here, y_j^i denotes the fraction of customers in size group i who visit option j in the second stage. We use index 0 to denote the no-purchase choice in both stages.

To complete the reformulation, we also introduce a vector of auxiliary variables $\vec{Q} := \{Q_1, \dots, Q_n\}$ where $Q_j := \{q_{j,0}, q_{j,1}, \dots, q_{j,n}\}$ for each $j \in N$. The role of $\vec{Q}$ is to impose the MNL structure in customers' choice in the second stage. More specifically, $q_{j,w} / (\sum_{m \in N} q_{j,m} + q_{j,0})$ denotes the fraction of customers who choose to visit option w from all customers who chose to revisit after visiting option j 's webpage. Therefore, we introduce the selection-based transformation of

(RBP-Assortment) as follows:

$$\begin{aligned}
\Pi_{\text{Revisit}}^{\text{Selection-Based}} &:= \max_{\psi, \mathbf{x}, \hat{Y}, \mathbf{u}, \hat{Q}} \sum_{i \in G} \xi^i \sum_{j \in N} (x_j + y_j^i) f^i(C_j) r_j - \psi \\
\text{s.t. } \quad &\psi \geq \lambda[u^l - u^k], \quad \forall k, l \in G, \\
&u^i = (x_0 + y_0^i) + \sum_{j \in N} (x_j + y_j^i)(1 - f^i(C_j)), \quad \forall i \in G, \\
&\frac{x_j}{v_j} \leq \frac{x_0}{v_0} \quad \forall j \in N, \quad \frac{q_{j,m}}{v_m} \leq \frac{q_{j,0}}{v_0} \quad j \in N, m \in N, \\
&\sum_{j \in N} x_j + x_0 = 1, \quad \sum_{m \in N} q_{j,m} + q_{j,0} = \delta x_j \quad \forall j \in N, \\
&y_m^i = \sum_{j \in N} q_{j,m} (1 - f^i(C_j)) \quad i \in G, m \in N \cup \{0\}, \\
&q_{m,j} \geq 0 \quad \forall m \in N, j \in N \cup \{0\}, \quad q_{j,j} = 0 \quad \forall j \in N, \quad x_j \geq 0 \quad \forall j \in N \cup \{0\},
\end{aligned}
\tag{RBP-Selection}$$

where (RBP-Selection) stands for *Revisit Behavior Problem - Selection-Based*. After solving this problem and obtaining $\hat{X}$, $\hat{Q}$, and $\hat{Y}$, we run a backward transformation similar to the one in the baseline model to obtain $\hat{\mathcal{P}}$. The details of the backward transformation stage are available in Appendix EC.3. We then prove the following result:

THEOREM 3. (i): *The optimal values of (RBP-Assortment) and (RBP-Selection) are equal.* (ii) *Allocation $\hat{\mathcal{P}}$ achieves the optimal solution of (RBP-Assortment).* (iii) *Expected revenue and unfairness are identical in the two formulations.*

To show the baseline model is a more constrained case of (RBP-Assortment), it suffices to add the constraint $\mathbf{P}_{2,(j)}(S_2 = \emptyset) = 1$ for every option $j \in N$. This represents a case where the retailer shows an empty assortment to customers who decided to revisit. Therefore, as in the size-filtering case, the optimal objective with revisits is higher than the baseline due to higher revenue, lower unfairness, or both. We examine the impact of this extension on the outcomes through extensive simulations in Section 6.3.

6. Managerial Insights

In this section, we use simulations calibrated to our industry partner's data to translate the model into managerial implications. We organize the analysis around four questions: how much fairness the retailer should pursue, which operational capabilities and customer behaviors improve the revenue-fairness trade-off, how a fairer display is implemented, and how sensitive these conclusions are to the market environment.

6.1. Simulation Setup

We calibrate the simulations using the dataset provided by our industry partner. We adapt the difference-in-differences approach of Li et al. (2023) to estimate size-substitution behavior;

the full specification, estimation procedure, and results are provided in Appendix EC.6. We use the resulting estimate, together with the industry data, to calibrate the distributions of option preference weights and revenues. In the baseline simulations, we set $p^u = p^l = 0.25$, $G = \{XS, S, M, L, XL, XXL\}$, and $n = 100$. The customer size distribution, option characteristics, and size-availability patterns mimic those in the data. Unless stated otherwise, results are averaged over 100 randomly generated inventory samples, and λ is reported as a fraction of $\max_j(r_j)$. When reporting normalized revenue and reductions in unfairness, we use the baseline revenue-maximizing policy ($\lambda = 0$) as the reference point.

6.2. How Much Fairness Should the Retailer Pursue?

We first examine the retailer's central economic trade-off: how much revenue must be sacrificed to reduce disparity in experienced size unavailability? Figure 4 traces the resulting revenue–fairness frontier. The cost of fairness is strongly nonlinear. At $\lambda = 0.1$, the retailer reduces unfairness by 26.27% while sacrificing only 1.29% of revenue. At $\lambda = 0.25$, unfairness falls by 59.69% while revenue declines by 6.91%. Pushing further to $\lambda = 0.5$ reduces unfairness by 86.09%, but the revenue loss increases to 16.14%. Thus, following directly from the concavity of the revenue–fairness frontier proved in Lemma 3, substantial early improvements are relatively inexpensive, whereas aggressive equalization becomes progressively more costly. Figure 5 sheds light on the intuition behind this by displaying experienced unavailability separately by size group. For small values of λ , our approach decreases unfairness mainly by reducing unavailability for the marginal size. However, for large values of λ , in contrast, the unfairness reduction is achieved in part by increasing unavailability among better-served groups.

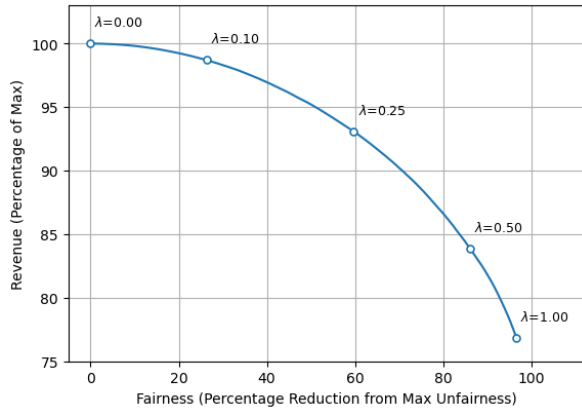


Figure 4 Revenue–fairness frontier, averaged over 100 inventory samples. λ is normalized with respect to $\max_j(r_j)$.

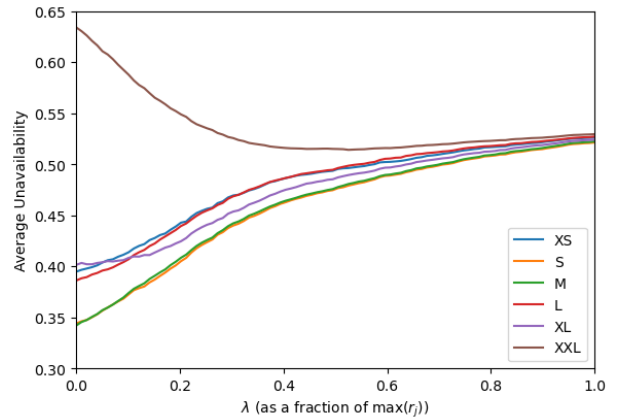


Figure 5 Experienced unavailability by size group, averaged over 100 inventory samples. λ is normalized with respect to $\max_j(r_j)$.

6.3. Use of Operational Capabilities in Improving the Outcomes

In the previous section, we observed that inducing fairness in the assortment can become costly, especially for large values of λ , where the retailer is forced to increase unavailability for mainstream sizes. We next examine how the two operational capabilities we introduced in Section 5 can help the retailer improve the outcome without incurring too much cost.

Model	$\lambda = 0.1$				$\lambda = 0.25$				$\lambda = 0.5$			
	Rev.	Unf. red.	$U_{\max}$	$U_{\min}$	Rev.	Unf. red.	$U_{\max}$	$U_{\min}$	Rev.	Unf. red.	$U_{\max}$	$U_{\min}$
Baseline	98.71 (0.72)	26.27 (11.87)	0.619 (0.052)	0.400 (0.025)	93.09 (2.05)	59.69 (13.57)	0.578 (0.054)	0.457 (0.030)	83.86 (5.02)	86.09 (6.61)	0.560 (0.046)	0.519 (0.042)
Size filtering	109.15 (1.46)	54.90 (15.19)	0.513 (0.059)	0.376 (0.019)	104.92 (2.09)	79.09 (16.11)	0.513 (0.059)	0.446 (0.025)	97.85 (7.08)	99.97 (0.31)	0.513 (0.059)	0.513 (0.059)
Revisit ($\delta = 0.5$) with revised assortment	105.06 (0.74)	34.13 (7.84)	0.564 (0.048)	0.367 (0.020)	101.12 (1.58)	57.13 (10.82)	0.541 (0.050)	0.413 (0.027)	91.14 (4.46)	85.03 (8.16)	0.529 (0.051)	0.484 (0.042)

Table 2 Effect of size filtering and customer revisits. Revenue is normalized to 100 at the baseline revenue-maximizing policy ($\lambda = 0$), and unfairness reduction is measured relative to the unfairness of that same policy. $U_{\max}$ and $U_{\min}$ denote the highest and lowest unavailability rates across size groups. Results are averaged over 100 inventory samples, with standard deviations in parentheses.

Table 2 compares the baseline model with size filtering and customer-revisit extensions. Size filtering directly reduces unequal exposure because the retailer observes the customer’s size before displaying the category page. By allowing the retailer to tailor the assortment to each size group, size filtering substantially reduces unavailability across all size groups, including the worst-off group, while greatly improving expected revenue. As the table shows, size filtering always offers higher revenue and lower unfairness for the same λ than the baseline, size-agnostic model.

Revisit opportunities operate differently. Size filtering acts directly on the exposure mechanism by using size information before an option is chosen. Revisit opportunities operate mainly after a mismatch has occurred and therefore primarily recover otherwise lost sales. This distinction is visible at $\lambda = 0.25$, where revisits raise revenue relative to the baseline while producing a slightly smaller reduction in unfairness. We also combine the two capabilities and develop a hybrid method available in the online appendix. Nonetheless, we observe that combining the two methods yields diminishing incremental gains in outcomes.

Taken together, these results highlight the distinct value of additional information and personalization. Size filtering can improve both revenue and fairness by using size information before an option is chosen, whereas revisit opportunities primarily recover revenue after an unavailable-size encounter. When both capabilities are available, combining them yields diminishing incremental gains.

6.4. How Is a Fairer Display Implemented?

We next examine what the retailer changes in the displayed assortment. Figure 6 reports three features of the optimal policy: the number of assortments in its randomized support, the number

of options displayed in at least one of those assortments, and the fraction of displayed options inherited from the revenue-maximizing assortment; we call these “revenue-maximizing options”. Taken together, the panels show that fairness is implemented primarily through *selective pruning and limited rotation*, rather than by complete deviation from the revenue-maximizing assortment. As λ increases, the retailer displays fewer options and randomizes over a larger, but still modest, set of complementary assortments. At the same time, most displayed options continue to belong to the revenue-maximizing core, with a limited set of options absent from the revenue-maximizing core, which we call the “peripheral options”.

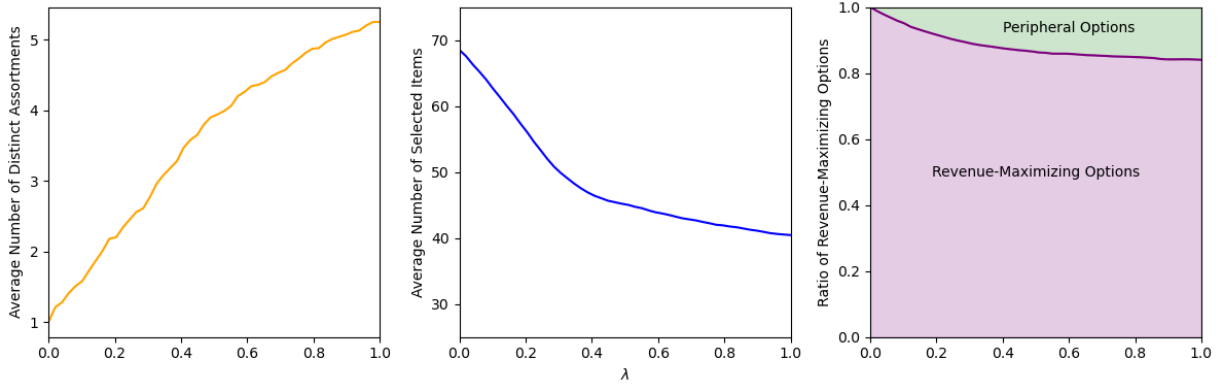


Figure 6 Implementation of the fairness-aware policy, averaged over 100 inventory samples. The left panel reports the support size, the middle panel the number of options displayed in at least one assortment, and the right panel the fraction of displayed options inherited from the revenue-maximizing assortment. λ is normalized with respect to $\max_j(r_j)$.

Table 3 further shows which revenue-maximizing options tend to be retained or suppressed. Conditional on our simulation design, in which option revenues are generated independently of size availability, retained revenue-maximizing options have systematically higher r_j than removed ones. For example, at $\lambda = 0.25$, the average r_j is 58.76 among retained options and 47.82 among removed options. The policy therefore tends to preserve commercially valuable options and concentrate more of the fairness adjustment on lower-revenue options when their size configurations permit it. Peripheral options are introduced selectively to balance size-group exposure.

6.5. Robustness to Market Environment

We finally examine how robust our methodology is to different problem parameters. We vary several primitives that can differ materially across retailers and categories: the shape of the customer size curve, the dispersion of option revenues and valuations, and the intensity of size-substitution behavior.

Table 4 compares concentrated, real-world, and uniform size distributions. The qualitative revenue–fairness trade-off is stable across the three cases, and the main managerial conclusion does

Option group	$\lambda = 0.1$	$\lambda = 0.25$	$\lambda = 0.5$
Revenue-maximizing options retained	57.13 (2.01)	58.76 (2.06)	59.05 (2.24)
Revenue-maximizing options removed	45.01 (3.37)	47.82 (2.84)	51.27 (2.69)
Peripheral options introduced	34.35 (10.99)	35.38 (7.25)	36.56 (7.46)

Table 3 Average option revenue r_j for options retained, removed, or introduced by the fairness-aware policy. Results are averaged over 100 inventory samples, with standard deviations in parentheses. λ is normalized with respect to $\max_j(r_j)$.

not depend on whether demand is concentrated in a few sizes or spread more evenly across the size range. Table 5 provides a more demanding robustness test. When valuations and/or option revenues are concentrated, increasing λ continues to generate large fairness improvements under both revenue distributions. This impact, however, is weaker for dispersed distributions. Considering our simulation assumption (where size stockouts are independent of option valuation or revenue), this can be explained through substitutability. More concentrated valuation/revenue distributions mean options are easier to substitute. In contrast, more dispersed distributions mean costlier substitutions.

Size-curve shape	$\lambda = 0.1$		$\lambda = 0.25$		$\lambda = 0.5$	
	Revenue	Unf. red.	Revenue	Unf. red.	Revenue	Unf. red.
Concentrated	98.95 (0.60)	19.94 (8.83)	93.64 (2.03)	50.75 (12.28)	82.14 (4.31)	83.18 (7.48)
Real-world	98.86 (0.59)	22.84 (10.38)	93.15 (2.49)	56.17 (14.09)	82.45 (5.13)	86.45 (8.85)
Uniform	98.75 (0.65)	25.30 (10.88)	92.99 (2.53)	59.33 (14.57)	83.20 (6.46)	85.93 (9.06)

Table 4 Impact of the shape of the customer size-curve on the fairness-aware policy. Revenue and unfairness reduction are measured relative to the revenue-maximizing policy ($\lambda = 0$). Results are averaged over 100 inventory samples, with standard deviations in parentheses.

Revenue dispersion	Value dispersion	$\lambda = 0.1$		$\lambda = 0.25$		$\lambda = 0.5$	
		Revenue	Unf. red.	Revenue	Unf. red.	Revenue	Unf. red.
Concentrated	Concentrated	98.90 (0.42)	20.56 (6.87)	91.88 (2.04)	58.32 (14.70)	79.94 (4.68)	93.79 (4.45)
Concentrated	Dispersed	99.36 (2.27)	9.21 (19.70)	94.96 (13.44)	22.75 (30.72)	73.50 (28.59)	55.27 (35.65)
Uniform	Concentrated	99.10 (0.40)	19.79 (7.11)	94.67 (1.69)	48.43 (10.85)	83.79 (3.34)	81.52 (9.08)
Uniform	Dispersed	94.98 (16.23)	15.79 (29.79)	84.80 (25.14)	35.19 (37.38)	71.12 (30.80)	53.61 (37.91)

Table 5 Performance of the fairness-aware policy under different revenue and valuation dispersions. Revenue and unfairness reduction are measured relative to the revenue-maximizing policy ($\lambda = 0$). Results are averaged over 100 inventory samples, with standard deviations in parentheses.

Finally, Table 6 summarizes the impact of size substitutions on the effectiveness of our method. We interpret the results by focusing on how size substitution works. With higher levels of sub-

stitution, an unavailable SKU is less detrimental because customers are willing to substitute it with a neighboring size. However, without size substitution, each unavailable SKU is costly in terms of both revenue and fairness. This, in turn, makes the retailer more cautious in building its assortments, leading to a narrower assortment.

Substitution probabilities	$\lambda = 0.1$		$\lambda = 0.25$		$\lambda = 0.5$	
	# options	Core ratio	# options	Core ratio	# options	Core ratio
$p^u = p^l = 0$	65.41 (4.94)	0.941 (0.027)	52.75 (4.77)	0.905 (0.043)	45.69 (5.04)	0.867 (0.059)
$p^u = p^l = 0.25$	70.41 (4.54)	0.961 (0.023)	58.28 (4.82)	0.933 (0.033)	48.09 (5.57)	0.906 (0.046)
$p^u = p^l = 0.5$	73.04 (4.41)	0.973 (0.020)	64.96 (4.94)	0.945 (0.029)	53.25 (4.77)	0.923 (0.037)

Table 6 Effect of size-substitution propensity on the displayed assortment. “# options” is the number of options displayed in at least one assortment, and “Core ratio” is the fraction of displayed options that also belong to the revenue-maximizing ($\lambda = 0$) assortment. Results are averaged over 100 inventory samples, with standard deviations in parentheses.

Overall, the simulations yield four managerial conclusions. First, meaningful reductions in disparity can be achieved at relatively modest revenue cost, whereas pushing toward near-equalization becomes increasingly expensive and relies more heavily on leveling the experiences of better-served size groups. Second, complementary operational capabilities act through different channels. Size filtering prevents unequal exposure by using customers’ size information before an option is chosen; revisit opportunities primarily recover customers after an unavailable-size encounter. Third, the fairness-aware policy does not require a wholesale redesign of the assortment. It operates primarily through selective pruning of revenue-maximizing options and limited rotation among a modest set of complementary assortments. Finally, although the qualitative revenue–fairness trade-off is robust across different customer size distributions, the effect of a given fairness weight becomes more sensitive to the market environment as option characteristics become more heterogeneous. Accordingly, rather than prescribing a universal value of λ , our results suggest that retailers should construct and evaluate their own revenue–fairness frontier using category-specific demand, inventory, and option characteristics.

7. Discussion and Conclusion

Size non-inclusivity in fashion retail is partly an inventory problem, but once the selling season is underway it is also a display problem. Upstream planning, purchasing, allocation, and demand realization determine which size–option combinations remain physically available. The retailer’s in-season assortment decision cannot restore a missing SKU, but it determines how customers are exposed to the resulting broken size runs. We study this second, controllable stage: given an

inherited inventory with asymmetric size configurations, how should an online retailer manage its displayed assortment to balance revenue and disparities in experienced size unavailability?

We develop a two-stage choice model in which customers first choose an option and subsequently evaluate its size availability, allowing for substitution to neighboring sizes. We measure unfairness as the gap between the highest and lowest unavailability rates across size groups and incorporate this measure into the retailer’s fairness-adjusted revenue. The resulting randomized assortment problem is combinatorial in its original form. We adapt the selection-based approach to transform it into a polynomial-size linear program and provide a backward mapping that recovers an optimal randomized assortment policy. The equivalence of the two formulations makes the method exact while preserving an assortment-based implementation. We further extend the framework to two practically relevant settings: size filtering, where the retailer observes the customer’s size before displaying the category page, and option substitution, where a customer may revisit the category page after encountering an unavailable size.

The analysis clarifies both when fairness concerns matter and how the retailer responds to them. When all relevant options have complete size configurations, the revenue-maximizing assortment is also fair across size groups; the revenue–fairness tension emerges from broken and asymmetric size runs. Once such configurations are present, a fairness-aware policy can reduce disparity through two channels: *lifting*, by improving the experience of initially worse-served groups, and *leveling*, by reducing advantages enjoyed by better-served groups. Our numerical analysis, calibrated using data from our industry partner, shows that meaningful reductions in unfairness are available at modest revenue cost, while aggressive equalization becomes increasingly expensive and relies more heavily on leveling. Operational levers such as size filtering and tailoring the assortment for revisiting customers can help reduce reliance on the leveling channel in those cases. Size filtering can reduce unequal exposure before customers choose an option, whereas revisit opportunities primarily recover revenue after an unavailable-size encounter. Operationally, the retailer does not need to replace its commercial assortment wholesale. Fairer policies are implemented mainly through selective pruning of revenue-maximizing options and limited rotation among a modest number of complementary assortments. More broadly, the simulations show that the policy’s qualitative role is robust to different size-curve shapes, although the effect of a fixed fairness weight becomes more environment-dependent when option valuations are highly heterogeneous. The relevant managerial object is therefore the retailer-specific revenue–fairness frontier, rather than a universal value of λ .

Our model is deliberately focused on the in-season display decision and should be viewed as complementary to, rather than a substitute for, better upstream planning. A natural next step is to integrate the display problem with decisions that determine the inventory state itself, including size-curve estimation, pre-pack design, purchasing, allocation, and replenishment. Another

direction is to enrich customer search behavior by allowing customers to form consideration sets containing multiple options before evaluating size availability, or by replacing the MNL model with richer choice specifications. The online appendix develops additional variants of the framework that illustrate its flexibility beyond the two main extensions retained in the paper. Taken together, our results show that even when the retailer cannot repair an imperfect size inventory, it can still manage how the consequences of that inventory are distributed across customers, often without giving up much of the assortment's commercial value.

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Electronic Companion to “Designing Size Inclusive Fashion Assortments”

This electronic companion contains supporting derivations, proofs, and extensions for the main paper. Section EC.1 provides the continuous physical-size microfoundation for the configuration reaction coefficients, Section EC.2 contains the proofs of the main analytical results, Section EC.3 develops the option-substitution model, Section EC.4 combines option substitution with size filtering, and Section EC.5 considers endogenous size-configuration decisions. Section EC.6 describes the empirical estimation and simulation calibration.

EC.1. Choice Model: The Second Stage

We assume a customer’s physical size b follows the distribution $\eta(b)$. We define the ‘fit quality’ as a function $\text{fit}(b, i)$ to show the fit quality that a size- b customer assigns to a size- i SKU, whereby customers only consider sizes with positive fit quality. For instance, consider the function $\text{fit}(b, i) = 1 - |b - i|/t$ that constitutes a symmetric fit quality function for a size- b customer for a size i SKU with distance threshold t . With this function, the customer only considers SKU sizes with $|b - i| \leq t$.

In the second stage, upon observing C_j , the size- b customer defines her size consideration set $Z(b, C_j)$ as the set of sizes that are (i) available in C_j , and (ii) provide a positive fit quality. We assume each customer chooses the SKU size in $Z(b, C_j)$ with the highest fit quality (and nothing if $Z(b, C_j)$ is empty), and define the function $\pi(b, C_j)$ to denote the SKU size with the highest fit quality in $Z(b, C_j)$. The size of $\pi(b, C_j)$, then, is:

$$|\pi(b, C_j)| = \begin{cases} 1, & \text{if } Z(b, C_j) \text{ is not empty;} \\ 0, & \text{if } Z(b, C_j) \text{ is empty.} \end{cases}$$

As a customer purchases the SKU $\pi(b, C_j)$, the probability that a size- b customer chooses option j and then buys an SKU from it is:

$$P(j|S, b) = \frac{v_j |\pi(b, C_j)|}{v_0 + \sum_{k \in S} v_k}. \quad (\text{EC.1})$$

So, if a retailer offers assortment S , its expected revenue from a size- b customer is:

$$\text{Rev}(S|b) = \frac{\sum_{j \in S} v_j |\pi(b, C_j)| r_j}{v_0 + \sum_{k \in S} v_k}. \quad (\text{EC.2})$$

The retailer’s expected revenue from assortment S then is:

$$\text{Rev}(S) = \int_0^\infty \text{Rev}(S|b) \eta(b) db = \int_0^\infty \frac{\sum_{j \in S} v_j |\pi(b, C_j)| r_j}{v_0 + \sum_{k \in S} v_k} \eta(b) db. \quad (\text{EC.3})$$

We make the following assumptions to develop a closed-form formulation for (EC.3). First, SKU sizes are discrete numbers between 1 and g , that is, $G = \{1, 2, \dots, g\}$. Second, the function strictly

decreases as the distance between the customer size b and the SKU size- i increases. Third, the fit function is homogeneous to a shift, meaning $\text{fit}(b+x, i+x) = \text{fit}(b, i)$ for any $x \geq 0$. Fourth, for now, we assume $\text{fit}(b, i) < 0$ for $|i - b| > 1$, which implies that there are at most two sizes that provide an acceptable fit for a customer.

By the second assumption, there exists a unique $\sigma \in [0, 1]$ such that $\text{fit}(i+\sigma, i) = \text{fit}(i+\sigma, i+1)$ for any $i \in G$, meaning a customer with size $i+\sigma$ assigns the same fit quality to SKU sizes i and $i+1$. We can then think of ‘size group i ’ as customers whose size is in the interval $m^i := [i-1+\sigma, i+\sigma]$. For these customers, the primary size is i . If $i \in C_j$, then $\pi(b, C_j) = i$ for these customers because by the second assumption $\text{fit}(b, i) \geq \text{fit}(b, i-1)$ and $\text{fit}(b, i) \geq \text{fit}(b, i+1)$. These customers constitute a fraction $\xi^i := \int_{i-1+\sigma}^{i+\sigma} \eta(b)db$ of all customers. We define $\Xi := \{\xi^1, \dots, \xi^g\}$ as the fraction of customers in each size group. Figure EC.1a illustrates an example of customer size groups.

Beyond σ , there are two additional key values. There exists a unique $\sigma^+ \in [0, \sigma]$ such that $Z(b, G) = \{i, i+1\}$ for $b \in [i+\sigma^+, i+\sigma)$ and $Z(b, G) = \{i\}$ for $b \in [i, i+\sigma^+]$. In other words, for size group i customers whose size lies within $[i+\sigma^+, i+\sigma)$, size $i+1$ is included as the second option in $Z(b, G)$. Similarly, a unique value $\sigma^- \in [\sigma, 1]$ exists where $Z(b, G) = \{i, i-1\}$ for $b \in [i-1+\sigma, i-1+\sigma^-]$ and $Z(b, G) = \{i\}$ for $b \in [i-1+\sigma^-, i)$. Figure EC.1b provides an example illustrating the proper-fit sets for customers.

We assume $\eta(b)$ is constant within every size group. So, we can replace $\eta(b)$ for $b \in m^i$ with ξ^i . Although this assumption is not necessary to establish our theoretical results, we explain later how it significantly reduces the amount of data required to estimate the model’s coefficients. We can now rewrite (EC.3) as:

$$\text{Rev}(S) = \sum_{i=1}^g \int_{i-1+\sigma}^{i+\sigma} \text{Rev}(S | b) \eta(b) db = \sum_{i=1}^g \int_{i-1+\sigma}^{i+\sigma} \sum_{j \in S} \frac{v_j |\pi(b, C_j)| r_j}{v_0 + \sum_{k \in S} v_k} \eta(b) db.$$

The only term in $\text{Rev}(S)$ that depends on b is $\pi(b, C_j)$. Because $\eta(b)$ is constant for $b \in m^i$, we replace it with the constant ξ^i , and write $\sum_{i=1}^g \int_{i-1+\sigma}^{i+\sigma} |\pi(b, C_j)| \eta(b) db = \sum_{i=1}^g \xi^i \int_{i-1+\sigma}^{i+\sigma} |\pi(b, C_j)| db$. Now, depending on C_j , there are five possible cases:

$$\int_{i-1+\sigma}^{i+\sigma} |\pi(b, C_j)| db = \begin{cases} \int_{i-1+\sigma}^{i+\sigma} 1 db = 1, & \text{if } i \in C_j; \\ \int_{i+\sigma^+}^{i+\sigma} 1 db = \sigma - \sigma^+, & \text{if } i, i-1 \notin C_j \text{ but } i+1 \in C_j; \\ \int_{i-1+\sigma}^{i-1+\sigma^-} 1 db = \sigma^- - \sigma, & \text{if } i, i+1 \notin C_j \text{ but } i-1 \in C_j; \\ \int_{i+\sigma^+}^{i+\sigma} 1 db + \int_{i-1+\sigma}^{i-1+\sigma^-} 1 db = \sigma^- - \sigma^+, & \text{if } i \notin C_j \text{ but } \{i-1, i+1\} \subseteq C_j; \\ 0, & \text{o.w.} \end{cases} \quad (\text{EC.4})$$

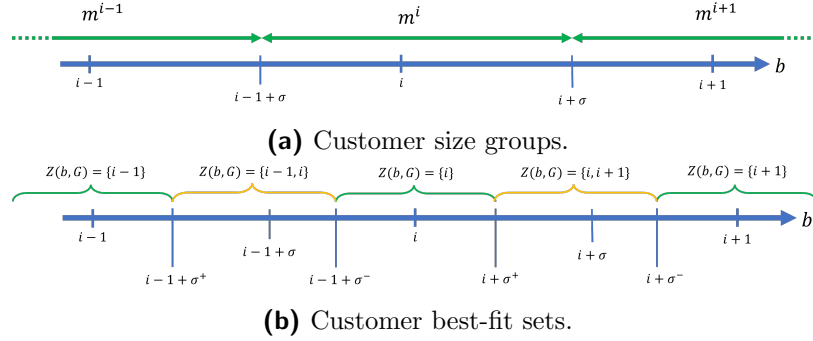


Figure EC.1 Illustration of customer size groups (m^i) and best-fit sets ($Z(b, G)$).

We define $p^u := \sigma - \sigma^+$ and $p^l := \sigma^- - \sigma$, which implies $p^u + p^l = \sigma^- - \sigma^+$. As such, we define the fraction of customers in size group i who select an SKU from option j when facing configuration C_j as *configuration reaction coefficient*, and denote it with $f^i(C_j)$ where:

$$f^i(C_j) := \int_{i-1+\sigma}^{i+\sigma} |\pi(b, C_j)| db = \begin{cases} 1, & \text{if } i \in C_j; \\ p^u, & \text{if } i, i-1 \notin C_j \text{ but } i+1 \in C_j; \\ p^l, & \text{if } i, i+1 \notin C_j \text{ but } i-1 \in C_j; \\ p^u + p^l, & \text{if } i \notin C_j \text{ but } \{i-1, i+1\} \subseteq C_j; \\ 0, & \text{o.w.} \end{cases} \quad (\text{EC.5})$$

To end this section, we explain why $\eta(b)$ being constant within every size group is helpful. Without this assumption, configuration reaction coefficients differ across size groups. This increases the number of coefficients from two (p^u and p^l) to $2g$ (two coefficients for every size group), significantly increasing the required data for estimation.

EC.2. Proofs

Proof of Lemma 1 Please refer to Appendix EC.1.

Proof of Lemma 2 For a given $\mathbf{P}$, we have $Unf(\mathbf{P}) = \max_{l,k \in G} \{U^l(\mathbf{P}) - U^k(\mathbf{P})\}$. So, we must have $Unf(\mathbf{P}) \geq U^l(\mathbf{P}) - U^k(\mathbf{P})$ and $Unf(\mathbf{P}) \geq U^k(\mathbf{P}) - U^l(\mathbf{P})$, meaning that $Unf(\mathbf{P}) \geq 0$ for any $\mathbf{P}$.

Take λ_1 and λ_2 such that $0 < \lambda_1 < \lambda_2$. We show the optimal assortment for λ_1 and λ_2 with $\mathbf{P}_1$ and $\mathbf{P}_2$, respectively. We have:

$$\begin{aligned} \Pi(\mathbf{P}_1) &= Rev(\mathbf{P}_1) - \lambda_1 Unf(\mathbf{P}_1) \geq Rev(\mathbf{P}_2) - \lambda_1 Unf(\mathbf{P}_2) \\ &\geq Rev(\mathbf{P}_2) - \lambda_2 Unf(\mathbf{P}_2) = \Pi(\mathbf{P}_2), \end{aligned}$$

where the first inequality holds because $\mathbf{P}_1$ is the optimal solution for $\lambda = \lambda_1$, and the second inequality comes from the fact that $Unf(\mathbf{P}_2) \geq 0$. This proves $\Pi(\mathbf{P}_1) \geq \Pi(\mathbf{P}_2)$. For the revenue and unfairness terms, from the optimality of $\mathbf{P}_1$ and $\mathbf{P}_2$, we write:

$$\begin{aligned} Rev(\mathbf{P}_1) - \lambda_1 Unf(\mathbf{P}_1) &\geq Rev(\mathbf{P}_2) - \lambda_1 Unf(\mathbf{P}_2), \\ Rev(\mathbf{P}_2) - \lambda_2 Unf(\mathbf{P}_2) &\geq Rev(\mathbf{P}_1) - \lambda_2 Unf(\mathbf{P}_1). \end{aligned}$$

Define $\Delta Rev := Rev(\mathbf{P}_1) - Rev(\mathbf{P}_2)$ and $\Delta Unf := Unf(\mathbf{P}_1) - Unf(\mathbf{P}_2)$. We have:

$$\Delta Rev \geq \lambda_1 \Delta Unf,$$

$$\Delta Rev \leq \lambda_2 \Delta Unf.$$

There are four possibilities for the sign of ΔRev and ΔUnf . Suppose they are both strictly negative. Then $|\Delta Rev| \leq \lambda_1 |\Delta Unf|$ contradicts $|\Delta Rev| \geq \lambda_2 |\Delta Unf|$, because $\lambda_2 > \lambda_1$. Also, if only one of them is strictly negative, then multiplying by λ_1 or λ_2 should not change the direction of inequality, meaning this possibility is infeasible. So, the only feasible possibility is $\Delta Rev \geq 0$ and $\Delta Unf \geq 0$. One can easily extend the proof to (AOP-DetFair) by replacing $\mathbf{P}^i$ with S^i , the optimal single assortment for λ_i . $\square$

Proof of Lemma 3 Take any two distributions $\mathbf{P}_1$ and $\mathbf{P}_2$. We have the following:

$$\begin{aligned} Rev(\alpha \mathbf{P}_1 + (1 - \alpha) \mathbf{P}_2) &= \sum_{w=1}^{2^n} (\alpha P_1(S_w) + (1 - \alpha) P_2(S_w)) Rev(S_w) \\ &= \alpha \sum_{w=1}^{2^n} P_1(S_w) Rev(S_w) + (1 - \alpha) \sum_{w=1}^{2^n} P_2(S_w) Rev(S_w) = \alpha Rev(\mathbf{P}_1) + (1 - \alpha) Rev(\mathbf{P}_2). \end{aligned}$$

So, the revenue term is linear in $\mathbf{P}$. Similarly, we can prove the same for each of the unavailability terms. For the unfairness term, we have:

$$\begin{aligned} Unf(\alpha \mathbf{P}_1 + (1 - \alpha) \mathbf{P}_2) &= \max_{l,k \in G} [U^k(\alpha \mathbf{P}_1 + (1 - \alpha) \mathbf{P}_2) - U^l(\alpha \mathbf{P}_1 + (1 - \alpha) \mathbf{P}_2)] \\ &= \max_{l,k \in G} [\alpha U^k(\mathbf{P}_1) + (1 - \alpha) U^k(\mathbf{P}_2) - \alpha U^l(\mathbf{P}_1) - (1 - \alpha) U^l(\mathbf{P}_2)] \\ &\leq \alpha \max_{l,k \in G} [U^k(\mathbf{P}_1) - U^l(\mathbf{P}_1)] + (1 - \alpha) \max_{l,k \in G} [U^k(\mathbf{P}_2) - U^l(\mathbf{P}_2)], \end{aligned}$$

which proves the convexity of $Unf(\mathbf{P})$ in $\mathbf{P}$.

As we mentioned, we define $Fair(\mathbf{P})$ as the complement of $Unf(\mathbf{P})$. Therefore, $Fair(\cdot)$ is concave in $\mathbf{P}$. For every $\lambda \geq 0$, we define $\mathbf{P}(\lambda)$ as the optimal solution for the given λ . Take any two values $\lambda_1 < \lambda_2$ where $(Rev(\mathbf{P}(\lambda_1)), Fair(\mathbf{P}(\lambda_1)))$ and $(Rev(\mathbf{P}(\lambda_2)), Fair(\mathbf{P}(\lambda_2)))$ are distinct. For any $\lambda \in (\lambda_1, \lambda_2)$ with a $(Rev(\mathbf{P}(\lambda)), Fair(\mathbf{P}(\lambda)))$ distinct from the past two pairs, Lemma 2 implies that $Rev(\mathbf{P}(\lambda_2)) < Rev(\mathbf{P}(\lambda)) < Rev(\mathbf{P}(\lambda_1))$ and $Fair(\mathbf{P}(\lambda_2)) > Fair(\mathbf{P}(\lambda)) > Fair(\mathbf{P}(\lambda_1))$. We find the unique α for which $Rev(\mathbf{P}(\lambda)) = \alpha Rev(\mathbf{P}(\lambda_1)) + (1 - \alpha) Rev(\mathbf{P}(\lambda_2))$. Suppose $Fair(\mathbf{P}(\lambda)) < \alpha Fair(\mathbf{P}(\lambda_1)) + (1 - \alpha) Fair(\mathbf{P}(\lambda_2))$. From the linearity of $Rev(\mathbf{P})$ and concavity of $Fair(\mathbf{P})$ in $\mathbf{P}$, we know we can use $\mathbf{P}' = \alpha \mathbf{P}(\lambda_1) + (1 - \alpha) \mathbf{P}(\lambda_2)$ and have the same revenue as $Rev(\mathbf{P}(\lambda))$, but a strictly larger fairness, meaning strictly lower unfairness. This contradicts the optimality of $\mathbf{P}(\lambda)$. Therefore, the revenue-fairness frontier is concave with respect to the origin. $\square$

Proof of Lemma 4 We first prove S^{nb} maximizes the revenue term, $Rev(S)$, in (AOP-DetFair). If $C_j = G$, then $\tilde{r}_j = r_j$. Therefore, we have $\tilde{r}_j = r_j$ for all $j \in N$. So the optimal assortment must maximize $(\sum_{j \in S} v_j r_j) / (v_0 + \sum_{k \in S} v_k)$, which is the set S^{nb} . Therefore, assortment (S^{nb}) maximizes $Rev(S)$ in (AOP-DetFair). Also, because $U^i(S^{nb}) = v_0 / (v_0 + \sum_{k \in S^{nb}} v_k)$ for every size group i ,

it follows that $\text{Unf}(S^{nb}) = 0$, which by Lemma 2 is the lower bound for the unfairness term. So assortment S^{nb} maximizes $\Pi(S) = \text{Rev}(S) - \lambda \text{Unf}(S)$. Without a broken assortment, (S^{nb}) is feasible; hence, it is the optimal assortment. $\square$

Proof of Theorem 1 Take $(\hat{\mathbf{x}}, \hat{\psi})$ as the optimal solution to (AOP-Selection). We construct $\hat{\mathbf{P}}$ as instructed in (13). First, we have the following for every option $j \in N \cup \{0\}$:

$$\Pr(\text{Select } j) = \sum_{k=0}^n \hat{P}(S_k) \Pr(j|S_k) = \sum_{k=j}^n \left(\frac{\hat{x}_k}{v_k} - \frac{\hat{x}_{k+1}}{v_{k+1}} \right) \left[v_0 + \sum_{m \in S_k} v_m \right] \frac{v_j}{v_0 + \sum_{m \in S_k} v_m} = \hat{x}_j.$$

Next, note that:

$$\begin{aligned} \sum_{S \subseteq N} \hat{P}(S) &= \sum_{j=0}^n \hat{P}(S_j) = \sum_{j=0}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) \left[v_0 + \sum_{m \in S_j} v_m \right] = v_0 \sum_{j=0}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) + \sum_{j=1}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) \sum_{m=1}^j v_m \\ &= \hat{x}_0 + \sum_{m=1}^n v_m \sum_{j=m}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) = \hat{x}_0 + \sum_{m=1}^n v_m \frac{\hat{x}_m}{v_m} = 1. \end{aligned}$$

As for the unavailability terms, we have:

$$\begin{aligned} \sum_{S \subseteq N} \hat{P}(S) \text{U}^i(s) &= \sum_{j=0}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) \left[v_0 + \sum_{m \in S_j} v_m \right] \left[\frac{v_0 + \sum_{m \in S_j} v_m (1 - f^i(C_m))}{v_0 + \sum_{m \in S_j} v_m} \right] \\ &= v_0 \sum_{j=0}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) + \sum_{j=1}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) \sum_{m=1}^j v_m (1 - f^i(C_m)) \\ &= \hat{x}_0 + \sum_{m=1}^n v_m (1 - f^i(C_m)) \sum_{j=m}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) = \hat{x}_0 + \sum_{m=1}^n \hat{x}_m (1 - f^i(C_m)). \end{aligned}$$

Therefore, for each unavailability pair, we have:

$$\sum_{S \subseteq N} \hat{P}(S) [\text{U}^k(S) - \text{U}^l(S)] = \sum_{j \in N} \hat{x}_j (1 - f^k(C_j)) - \sum_{j \in N} \hat{x}_j (1 - f^l(C_j)) = \sum_{j \in N} \hat{x}_j (f^l(C_j) - f^k(C_j)).$$

Therefore, by utilizing $\hat{\mathbf{P}}$, the two sets of unfairness constraints in (AOP-Assortment) and (AOP-Selection) admit the same upper bound. Similar to the unfairness terms, we can write the revenue term:

$$\begin{aligned} \sum_{S \subseteq N} \hat{P}(S) \text{Rev}(S) &= \sum_{j=0}^n \hat{P}(S_j) \text{Rev}(S_j) = \sum_{j=0}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) \left[v_0 + \sum_{m \in S_j} v_m \right] \frac{\sum_{m \in S_j} v_m \tilde{r}_m(C_m)}{v_0 + \sum_{m \in S_j} v_m} \\ &= \sum_{j=1}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) \sum_{m=1}^j v_m \tilde{r}_m(C_m) = \sum_{m=1}^n v_m \tilde{r}_m(C_m) \sum_{j=m}^n \left(\frac{\hat{x}_j}{v_j} - \frac{\hat{x}_{j+1}}{v_{j+1}} \right) = \sum_{m \in N} \hat{x}_m \tilde{r}_m(C_m). \end{aligned}$$

As such, $(\hat{\mathbf{P}}, \hat{\psi})$ is both feasible in (AOP-Assortment) and brings about the same objective value as $(\hat{\mathbf{x}}, \hat{\psi})$ does in (AOP-Selection). As such, we have that $\Pi^{\text{Assortment-Based}} \geq \Pi^{\text{Selection-Based}}$.

Suppose $(\mathbf{P}', \psi')$ is the optimal solution for (AOP-Assortment). We build a suggested solution for (AOP-Selection) by setting $x'_j := \sum_{S \subseteq N} P'(S) \Pr(j|S)$. It is easy to observe that $\sum_{j \in N} x'_j + x'_0 = 1$ and $x'_j/v_j \leq x'_0/v_0$. Now, we re-write the unfairness constraints with $\mathbf{x}'$:

$$\begin{aligned} \sum_{j \in N} x'_j (f^k(C_j) - f^l(C_j)) &= \sum_{j \in N} \left(\sum_{S \subseteq N} P'(S) \Pr(j|S) \right) [f^k(C_j) - f^l(C_j)] = \sum_{S \subseteq N} P'(S) \sum_{j \in N} \Pr(j|S) [f^k(C_j) - f^l(C_j)] \\ &= \sum_{S \subseteq N} P'(S) \sum_{j \in S} \frac{v_j}{v_0 + \sum_{m \in S} v_m} [f^k(C_j) - f^l(C_j)] \\ &= \sum_{S \subseteq N} P'(S) \left[\frac{v_0 + \sum_{j \in S} v_j (1 - f^l(C_j))}{v_0 + \sum_{m \in S} v_m} - \frac{v_0 + \sum_{j \in S} v_j (1 - f^k(C_j))}{v_0 + \sum_{m \in S} v_m} \right] \\ &= \sum_{S \subseteq N} P'(S) [U^l(S) - U^k(S)]. \end{aligned}$$

Therefore, by utilizing $\mathbf{x}'$, the two sets of unfairness constraints in (AOP-Assortment) and (AOP-Selection) admit the same upper bound. Similar to the unfairness terms, we can write the revenue term:

$$\begin{aligned} \sum_{j \in N} \tilde{r}_j(C_j) x'_j &= \sum_{j \in N} \left(\sum_{S \subseteq N} P'(S) \Pr(j|S) \right) \tilde{r}_j(C_j) = \sum_{S \subseteq N} P'(S) \sum_{j \in N} \Pr(j|S) \tilde{r}_j(C_j) \\ &= \sum_{S \subseteq N} P'(S) \sum_{j \in S} \frac{v_j \tilde{r}_j(C_j)}{v_0 + \sum_{m \in S} v_m} = \sum_{S \subseteq N} P'(S) \text{Rev}(S). \end{aligned}$$

Therefore, $(\mathbf{x}', \psi')$ is both feasible in (AOP-Selection) and brings about the same objective value as $(\mathbf{P}', \psi')$ does in (AOP-Assortment). As such, we have that $\Pi^{\text{Selection-Based}} \geq \Pi^{\text{Assortment-Based}}$.

Combining the two directions, we obtain that: (i) The optimal values of (AOP-Assortment) and (AOP-Selection) are equal. (ii) $\hat{\mathbf{P}}$ provides an optimal solution for (AOP-Assortment). (iii) Expected revenue and unfairness are identical in the two formulations. $\square$

Proof of Lemma 5 Because the objective function in (AOP-Selection) is bounded and its feasible space is non-empty, there exists a *basic feasible solution* for it (Chen et al. 2022), which we show with $(\hat{\mathbf{x}}, \hat{\psi})$. Suppose $\Pi^* > 0$ (otherwise, the optimal solution is $\hat{P}(\emptyset) = 1$, which is a singleton). If $\Pi^* > 0$, we must have $\hat{x}_j > 0$ for at least one option, meaning $\hat{x}_0 > 0$. Because there are $n + 2$ decision variables in (AOP-Selection) and $(\hat{\mathbf{x}}, \hat{\psi})$ is a basic feasible solution, at least $n + 2$ of the constraints must be active. Because $\sum_{j \in N} \hat{x}_j + \hat{x}_0 = 1$, at least $n + 1$ of the following constraints must be active:

$$\{\hat{\psi} \geq \lambda \sum_{j \in N} \hat{x}_j (f^k(C_j) - f^l(C_j)) \forall k, l \in G; \frac{\hat{x}_j}{v_j} \leq \frac{\hat{x}_0}{v_0} \forall j \in N; \hat{x}_j \geq 0 \forall j \in N\}.$$

At most g^2 of the first set of constraints are binding, meaning at least $n + 1 - g^2$ of the last two sets of constraints are binding. Note that the last two sets of constraints cannot be active at the same time for each option. That is, for any $j \in N$, $\hat{x}_j = 0$ and $\hat{x}_j/v_j = \hat{x}_0/v_0$ cannot be satisfied at the same time. Denote the total number of options that do not satisfy $\hat{x}_j = 0$ or $\hat{x}_j/v_j = \hat{x}_0/v_0$ with

p . The remaining $n - p$ options bind exactly $n - p$ constraints. Since $n + 1 - g^2$ is a lower bound to the number of binding constraints in the last two sets, we must have: $n - p \geq n + 1 - g^2 \Rightarrow p \leq g^2 - 1$.

Now, we interpret p . The way $\hat{\mathbf{P}}$ is constructed, options with the same $\hat{x}_j/v_j$ are grouped together and build a bundle. Suppose $\hat{x}_j/v_j = \hat{x}_{j+1}/v_{j+1} = \dots = \hat{x}_{j+l}/v_{j+l}$. The display probability assignment rule in (13) imposes that $\hat{P}(S_j) = \hat{P}(S_{j+1}) = \dots = \hat{P}(S_{j+l-1}) = 0$. Therefore, the number of bundles is equivalent to the number of distinct assortments that are displayed with a non-zero probability. The p options with $0 < \hat{x}_j/v_j < \hat{x}_0/v_0$ can create at most p bundles. The remaining $n - p$ options fall in one of the two bundles: bundle of options with $\hat{x}_j/v_j = \hat{x}_0/v_0$ or $\hat{x}_j = 0$. However, we can disregard options with $\hat{x}_j = 0$ because they do not create an assortment. Therefore, the number of assortments that are displayed at random with $\hat{\mathbf{P}}$ is at most $p + 1$ which is bounded by g^2 . Note that if $\hat{\psi} > 0$ in the optimal solution (meaning unfairness is not fully eradicated in the optimal solution), this limit can be further pushed down to $g^2/4$ for even g 's and $g(g-1)/4$ for odd g 's. $\square$

Proof of Theorem 2 The proof follows the same steps as the proof of Theorem 1. The only difference is that now we have a distinct set of display probabilities for each size group. Also, for the unavailability terms, we have $u^i(X^i) = x_0^i + \sum_{j \in N} x_j^i(1 - f^i(C_j))$. As such, for each unfairness pair we have:

$$\begin{aligned} u^k(X^k) - u^l(X^l) &= (x_0^k + \sum_{j \in N} x_j^k(1 - f^k(C_j))) - (x_0^l + \sum_{j \in N} x_j^l(1 - f^l(C_j))) \\ &= (x_0^k + \sum_{j \in N} x_j^k) - (x_0^l + \sum_{j \in N} x_j^l) + \sum_{j \in N} x_j^l f^l(C_j) - \sum_{j \in N} x_j^k f^k(C_j) = \sum_{j \in N} x_j^l f^l(C_j) - \sum_{j \in N} x_j^k f^k(C_j), \end{aligned}$$

where the last equality comes from the constraint that $x_0^i + \sum_{j \in N} x_j^i = 1$ for every size group $i \in G$. $\square$

Proof of Theorem 3 As in the baseline model, we construct $\hat{\mathbf{P}}_1$ based on $\hat{\mathbf{x}}$. We define the mapping σ so that $\hat{x}_{\sigma(1)}/v_{\sigma(1)} \geq \hat{x}_{\sigma(2)}/v_{\sigma(2)} \geq \dots \geq \hat{x}_{\sigma(n)}/v_{\sigma(n)}$. Also, for ease of notation, we define $\sigma(0)$ as the outside option and simply show it with 0, so $\hat{x}_0/v_0 \geq \hat{x}_{\sigma(1)}/v_{\sigma(1)}$. Based on that, nested sets $S_m := \{\sigma(1), \dots, \sigma(m)\}$ with $S_0 = \emptyset$ are defined. Setting $\hat{\mathbf{P}}_1(S) = 0$ for any assortment $S \notin \{S_1, \dots, S_n\}$ and $\hat{x}_{n+1} = 0$ we define: $\hat{P}_1(S_j) := \left(\frac{\hat{x}_{\sigma(j)}}{v_{\sigma(j)}} - \frac{\hat{x}_{\sigma(j+1)}}{v_{\sigma(j+1)}} \right) \left[v_0 + \sum_{i \in S_j} v_i \right]$ for $j = 0, 1, \dots, n$.

We build $\hat{\mathbf{P}}_{2,(j)}$'s based on $\hat{\mathbf{Q}}$ in the following way. We define the mapping σ_j so that $\hat{q}_{j,\sigma_j(1)}/v_{\sigma_j(1)} \geq \hat{q}_{j,\sigma_j(2)}/v_{\sigma_j(2)} \geq \dots \geq \hat{q}_{j,\sigma_j(n)}/v_{\sigma_j(n)}$. Also, for ease of notation, we define $\sigma_j(0)$ as the outside option and simply show it with 0, so $\hat{q}_{j,0}/v_0 \geq \hat{q}_{j,\sigma_j(1)}/v_{\sigma_j(1)}$. Based on that, nested sets $S_{(j),m} := \{\sigma_j(1), \dots, \sigma_j(m)\}$ with $S_{j,0} = \emptyset$ are defined. Setting $\hat{\mathbf{P}}_{2,(j)}(S) = 0$ for any assortment $S \notin \{S_{(j),1}, \dots, S_{(j),n}\}$ and $\hat{q}_{j,\sigma_j(n+1)} = 0$, we define:

$$\hat{P}_{2,(j)}(S_{(j),m}) := \frac{\left(\frac{\hat{q}_{j,\sigma_j(m)}}{v_{\sigma_j(m)}} - \frac{\hat{q}_{j,\sigma_j(m+1)}}{v_{\sigma_j(m+1)}} \right) \left[v_0 + \sum_{t \in S_{(j),m}} v_t \right]}{\hat{q}_{j,0} + \sum_{k \in N} \hat{q}_{j,\sigma_j(k)}} \quad \text{for } m = 0, 1, \dots, n. \quad (\text{EC.6})$$

We begin the proof by showing $(\hat{P}_1, \hat{P}_{2,(1)}, \dots, \hat{P}_{2,(n)})$ and $\hat{\psi}$ constitute a feasible solution for (RBP-Assortment). First, observe that

$$\begin{aligned} \sum_{S \subseteq N} \hat{P}_1(S) &= \sum_{j=0}^n \left(\frac{\hat{x}_{\sigma(j)}}{v_{\sigma(j)}} - \frac{\hat{x}_{\sigma(j+1)}}{v_{\sigma(j+1)}} \right) \left[v_0 + \sum_{i \in S_j} v_i \right] = v_0 \sum_{j=0}^n \left(\frac{\hat{x}_{\sigma(j)}}{v_{\sigma(j)}} - \frac{\hat{x}_{\sigma(j+1)}}{v_{\sigma(j+1)}} \right) + \sum_{j=1}^n \left(\frac{\hat{x}_{\sigma(j)}}{v_{\sigma(j)}} - \frac{\hat{x}_{\sigma(j+1)}}{v_{\sigma(j+1)}} \right) \sum_{t=1}^j v_{\sigma(t)} \\ &= \hat{x}_0 + \sum_{i=1}^n v_{\sigma(i)} \sum_{j=t}^n \left(\frac{\hat{x}_{\sigma(j)}}{v_{\sigma(j)}} - \frac{\hat{x}_{\sigma(j+1)}}{v_{\sigma(j+1)}} \right) = \hat{x}_0 + \sum_{i=1}^n v_{\sigma(i)} \frac{\hat{x}_{\sigma(i)}}{v_{\sigma(i)}} = 1, \end{aligned}$$

and

$$\begin{aligned} \sum_{S \subseteq N} \hat{P}_{2,(j)}(S) &= \frac{1}{\hat{q}_{j,0} + \sum_{k=1}^n \hat{q}_{j,\sigma_j(k)}} \sum_{m=0}^n \left(\frac{\hat{q}_{j,\sigma_j(m)}}{v_{\sigma_j(m)}} - \frac{\hat{q}_{j,\sigma_j(m+1)}}{v_{\sigma_j(m+1)}} \right) \left[v_0 + \sum_{t \in S_{(j),m}} v_t \right] \\ &= \frac{v_0}{\hat{q}_{j,0} + \sum_{k=1}^n \hat{q}_{j,\sigma_j(k)}} \sum_{m=0}^n \left(\frac{\hat{q}_{j,\sigma_j(m)}}{v_{\sigma_j(m)}} - \frac{\hat{q}_{j,\sigma_j(m+1)}}{v_{\sigma_j(m+1)}} \right) \\ &\quad + \frac{1}{\hat{q}_{j,0} + \sum_{k=1}^n \hat{q}_{j,\sigma_j(k)}} \sum_{m=0}^n \left(\frac{\hat{q}_{j,\sigma_j(m)}}{v_{\sigma_j(m)}} - \frac{\hat{q}_{j,\sigma_j(m+1)}}{v_{\sigma_j(m+1)}} \right) \sum_{t=1}^m v_{\sigma_j(t)} \\ &= \frac{\hat{q}_{j,0}}{\hat{q}_{j,0} + \sum_{k=1}^n \hat{q}_{j,\sigma_j(k)}} + \frac{1}{\hat{q}_{j,0} + \sum_{k=1}^n \hat{q}_{j,\sigma_j(k)}} \sum_{t=1}^n v_{\sigma_j(t)} \sum_{m=t}^n \left(\frac{\hat{q}_{j,\sigma_j(m)}}{v_{\sigma_j(m)}} - \frac{\hat{q}_{j,\sigma_j(m+1)}}{v_{\sigma_j(m+1)}} \right) \\ &= \frac{\hat{q}_{j,0}}{\hat{q}_{j,0} + \sum_{k=1}^n \hat{q}_{j,\sigma_j(k)}} + \frac{1}{\hat{q}_{j,0} + \sum_{k=1}^n \hat{q}_{j,\sigma_j(k)}} \sum_{t=1}^n v_{\sigma_j(t)} \frac{\hat{q}_{j,\sigma_j(t)}}{v_{\sigma_j(t)}} = 1. \end{aligned}$$

Also, because $\hat{q}_{j,j} = 0$, and because the assortments built through (EC.6) follow a nested structure, $\hat{P}_{2,(j)}(S') = 0$ if $j \in S'$. For the next step in the proof, we establish the equivalence of the first- and second-stage expected numbers of visits per customer for a sample option j between the two models. To see that, we begin with the first stage:

$$\begin{aligned} \text{Expected number of visits for option } j \text{ per-customer at the first stage} &= \sum_{S \subseteq N} \hat{P}_1(S) \Pr(j|S) \\ &= \sum_{m=0}^n \left(\frac{\hat{x}_{\sigma(m)}}{v_{\sigma(m)}} - \frac{\hat{x}_{\sigma(m+1)}}{v_{\sigma(m+1)}} \right) \left[v_0 + \sum_{i \in S_m} v_i \right] \frac{v_j}{v_0 + \sum_{i \in S_m} v_i} \mathbf{1}(j \in S_m) \\ &= \sum_{m=\sigma^{-1}(j)}^n \left(\frac{\hat{x}_{\sigma(m)}}{v_{\sigma(m)}} - \frac{\hat{x}_{\sigma(m+1)}}{v_{\sigma(m+1)}} \right) v_j = v_j \frac{\hat{x}_{\sigma(\sigma^{-1}(j))}}{v_{\sigma(\sigma^{-1}(j))}} = \hat{x}_j, \end{aligned} \tag{EC.7}$$

where $\sigma^{-1}(j)$ is the rank of option j in the mapping σ . Due to the nested structure of this mapping, for any $m \geq \sigma^{-1}(j)$, we have $j \in S_m$. We now move to establishing a similar result for the second

stage. We first need to establish an intermediary result:

$$\begin{aligned}
\sum_{S' \subseteq N} \hat{P}_{2,(e)}(S') \Pr(j|S') &= \sum_{m=0}^n \frac{\left(\frac{\hat{q}_{e,\sigma_e(m)}}{v_{\sigma_e(m)}} - \frac{\hat{q}_{e,\sigma_e(m+1)}}{v_{\sigma_e(m+1)}} \right) \left[v_0 + \sum_{t \in S_{(e),m}} v_t \right]}{\hat{q}_{e,0} + \sum_{k=1}^n \hat{q}_{e,\sigma_e(k)}} \frac{v_j}{v_0 + \sum_{t \in S_{(e),m}} v_t} \mathbf{1}(j \in S_{(e),m}) \\
&= \sum_{m=\sigma_e^{-1}(j)}^n \frac{\left(\frac{\hat{q}_{e,\sigma_e(m)}}{v_{\sigma_e(m)}} - \frac{\hat{q}_{e,\sigma_e(m+1)}}{v_{\sigma_e(m+1)}} \right) \left[v_0 + \sum_{t \in S_{(e),m}} v_t \right]}{\hat{q}_{e,0} + \sum_{k=1}^n \hat{q}_{e,\sigma_e(k)}} \frac{v_j}{v_0 + \sum_{t \in S_{(e),m}} v_t} \\
&= \frac{v_j}{\hat{q}_{e,0} + \sum_{k=1}^n \hat{q}_{e,\sigma_e(k)}} \frac{\hat{q}_{e,\sigma_e(\sigma_e^{-1}(j))}}{v_{\sigma_e(\sigma_e^{-1}(j))}} = \frac{\hat{q}_{e,j}}{\hat{q}_{e,0} + \sum_{k=1}^n \hat{q}_{e,\sigma_e(k)}} = \frac{\hat{q}_{e,j}}{\delta \hat{x}_e}.
\end{aligned} \tag{EC.8}$$

The definition of $\sigma_e^{-1}(j)$ and the reason behind the second equation are similar to that of (EC.7).

Now, with the use of (EC.7) and (EC.8), we develop the equivalence for the second stage.

$$\begin{aligned}
\text{Expected number of visits for option } j \text{ per-customer at second stage} &= \sum_{e \in N} \sum_{S' \subseteq N} \hat{P}_1(S) \Pr(e|S) \delta(1 - f^i(C_e)) \sum_{S' \subseteq N} \hat{P}_{2,(e)}(S') \Pr(j|S') \\
&= \sum_{e \in N} \hat{x}_e \delta(1 - f^i(C_e)) \frac{\hat{q}_{e,j}}{\delta \hat{x}_e} = \sum_{e \in N} \hat{q}_{e,j} (1 - f^i(C_e)) = \hat{y}_j^i.
\end{aligned} \tag{EC.9}$$

Now, combining (EC.7) and (EC.9), we have:

$$\text{EV}(j, i | \hat{\mathbf{P}}_1, \hat{\mathbf{P}}_{2,(1)}, \dots, \hat{\mathbf{P}}_{2,(n)}) = \hat{x}_j + \hat{y}_j^i \quad \forall j = 0, 1, \dots, n, i \in G.$$

For more details on the function $\text{EV}(\cdot)$, please refer to equation EC.10 in Appendix EC.3. Using this, it is straightforward to show the equivalence of the revenue and unavailability terms:

$$\begin{aligned}
E_{\hat{P}_1} \sum_{j \in N} E_{\hat{P}_{2,(j)}} \text{Rev}^i(S_1, S_2) &= \sum_{i \in G} \xi^i \sum_{j \in N} (\hat{x}_j + \hat{y}_j^i) f^i(C_j) r_j, \\
E_{P_1} \sum_{j \in N} E_{P_{2,(j)}} \text{U}^i(S_1, S_2) &= (\hat{x}_0 + \hat{y}_0^i) + \sum_{j \in N} (\hat{x}_j + \hat{y}_j^i) (1 - f^i(C_j)).
\end{aligned}$$

Therefore, $(\hat{\mathbf{P}}_1, \hat{\mathbf{P}}_{2,(1)}, \dots, \hat{\mathbf{P}}_{2,(n)})$ and $\hat{\psi}$ constitute a feasible solution for (RBP-Assortment), with the same revenue and level of unfairness. Now, we focus on the reverse direction. Suppose $(\mathbf{P}'_1, \mathbf{P}'_{2,(1)}, \dots, \mathbf{P}'_{2,(n)})$ and ψ' constitute an optimal solution for (RBP-Assortment). We build a suggested solution for (RBP-Selection) in the following way:

$$\begin{aligned}
x'_j &:= \sum_{S \subseteq N} P'_1(S) \Pr(j|S), \\
y_j^{i'} &:= \sum_{e \in N} \sum_{S' \subseteq N} P'_1(S) \Pr(e|S) \delta(1 - f^i(C_e)) \sum_{S' \subseteq N} P'_{2,(e)}(S') \Pr(j|S'), \\
q'_{e,j} &:= \delta x'_e \sum_{S' \subseteq N} P'_{2,(e)}(S') \Pr(j|S').
\end{aligned}$$

It is easy to show that $x'_j/v_j \leq x'_0/v_0$ and $q'_{j,m}/v_m \leq q'_{j,0}/v_0$ for all $j, m \in N$. Also, we know $q'_{j,j} = 0$ because $P'_{2,(j)}(S) = 0$ if $j \in S$. We then observe the following:

$$\begin{aligned} \sum_{j=0}^n x'_j &= \sum_{j=0}^n \sum_{S \subseteq N} P'_1(S) \Pr(j|S) = \sum_{S \subseteq N} P'_1(S) \sum_{j=0}^n \Pr(j|S) = \sum_{S \subseteq N} P'_1(S) = 1; \\ \sum_{j=0}^n q'_{e,j} &= \sum_{j=0}^n \delta x'_e \sum_{S' \subseteq N} P'_{2,(e)}(S') \Pr(j|S') = \delta x'_e \sum_{S' \subseteq N} P'_{2,(e)}(S') \sum_{j=0}^n \Pr(j|S') = \delta x'_e \sum_{S' \subseteq N} P'_{2,(e)}(S') = \delta x'_e; \\ \sum_{e \in N} q'_{e,j} (1 - f^i(C_e)) &= \sum_{e \in N} \delta x'_e \sum_{S' \subseteq N} P'_{2,(e)}(S') \Pr(j|S') (1 - f^i(C_e)) = \sum_{e \in N} \delta x'_e (1 - f^i(C_e)) \sum_{S' \subseteq N} P'_{2,(e)}(S') \Pr(j|S') \\ &= \sum_{e \in N} \sum_{S \subseteq N} P'_1(S) \Pr(e|S) \delta (1 - f^i(C_e)) \sum_{S' \subseteq N} P'_{2,(e)}(S') \Pr(j|S') = y_j^{i'}. \end{aligned}$$

Also, we have that

$$x'_j + y_j^{i'} = \text{EV}(j, i | \mathbf{P}'_1, \mathbf{P}'_{2,(1)}, \dots, \mathbf{P}'_{2,(n)}) \quad \forall j = 0, 1, \dots, n, i \in G.$$

It is straightforward to show the equivalence of the revenue and unavailability terms:

$$\begin{aligned} \sum_{i \in G} \xi^i \sum_{j \in N} (x'_j + y_j^{i'}) f^i(C_j) r_j &= \sum_{i \in G} \xi^i \sum_{j \in N} \text{EV}(j, i | \mathbf{P}', \mathbf{P}'_{-1}, \dots, \mathbf{P}'_{-n}) f^i(C_j) r_j, \\ (x'_0 + y_0^{i'}) + \sum_{j \in N} (x'_j + y_j^{i'}) (1 - f^i(C_j)) &= \text{EV}(0, i | \mathbf{P}', \mathbf{P}'_{-1}, \dots, \mathbf{P}'_{-n}) + \sum_{j \in N} \text{EV}(j, i | \mathbf{P}', \mathbf{P}'_{-1}, \dots, \mathbf{P}'_{-n}) (1 - f^i(C_j)). \end{aligned}$$

Therefore, $X', \vec{Y}', \vec{Q}'$ and ψ' constitute a feasible solution for (RBP-Selection), with the same expected revenue and unfairness terms. Combining the two directions, we see that the transformed optimal solution of each approach is feasible in the other approach and yields the same revenue and unfairness terms. $\square$

EC.3. Choice Model Subject to Option Substitution

Unlike physical stores, the online retailer is free to offer any assortment to customers who return to the category page. With the same logic, the retailer can also utilize randomization. Nevertheless, it is safe to assume that the customer will not select option j (her first choice) if it is present in the second-stage assortment. We introduce $N_{-j} := N \setminus \{j\}$ to denote the set of options from which the retailer may choose assortments to display to the returning customer.

As the customer has not yet visited the webpage of other options, we assume her valuation of options in N_{-j} has not changed. So, faced with $S \subseteq N_{-j}$, she will choose option e with probability $v_e/(v_0 + \sum_{m \in S} v_m)$. After visiting option e 's webpage, the size substitution behavior is similar to that developed in (2). As a reminder, the retailer chooses (i) a first-stage randomized assortment distribution $\mathbf{P}_1$, and (ii) second-stage randomized assortment distributions $\{\mathbf{P}_{2,(1)}, \dots, \mathbf{P}_{2,(n)}\}$, where $\mathbf{P}_{2,(j)}$ is shown only to customers who look for other options after they did not purchase option j . A modeling complexity here is that, with the possibility of revisits, the 'probability' of a customer visiting an option depends on the order of visits and differs between customers leaving

at stage one and those opting for a revisit. Therefore, given the possibility of revisits, we focus instead on the number of times an option is visited by each customer who enters the website. We define the ‘expected number of visits per customer’ (EV) in size group i in the following way:

$$\begin{aligned} \text{EV}(j, i | \mathbf{P}_1, \mathbf{P}_{2,(1)}, \dots, \mathbf{P}_{2,(n)}) = & \underbrace{\sum_{S_1 \subseteq N} P_1(S_1) \Pr(j | S_1)}_{\text{Expected number of visits per-customer at the first stage}} \\ & + \underbrace{\sum_{e \in N} \sum_{S_1 \subseteq N} P_1(S_1) \Pr(e | S_1) \delta(1 - f^i(C_e)) \sum_{S_2 \subseteq N} P_{2,(e)}(S_2) \Pr(j | S_2)}_{\text{Expected number of visits per-customer at the second stage}}. \end{aligned} \quad (\text{EC.10})$$

where $\Pr(j | S)$ follows the MNL structure defined in (1). The term for the first stage is straightforward, so we focus on the term for the second stage. Suppose a customer in size group i visited option e ’s webpage at the first stage. This happens to a fraction $\sum_{S_1 \subseteq N} P_1(S_1) \Pr(e | S_1)$ of entering customers. After that, a fraction $1 - f^i(C_e)$ of customers in size group i leave option e ’s webpage, and a fraction δ of them opt to revisit. These customers are shown an assortment drawn from distribution $\mathbf{P}_{2,(e)}$, so a fraction $\sum_{S_2 \subseteq N} P_{2,(e)}(S_2) \Pr(j | S_2)$ of them choose option j as their second choice. A fraction $f^i(C_j)$ of customers in size group i buy an SKU from option j after visiting its webpage, whether at the first or second stage. With this in mind, we write the revenue and unavailability in the revisit approach:

$$\begin{aligned} E_{P_1} \sum_{j \in N} E_{P_{2,(j)}} \text{Rev}^i(S_1, S_2) &:= \sum_{j \in N} \text{EV}(j, i | \mathbf{P}, \mathbf{P}_{-1}, \dots, \mathbf{P}_{-n}) f^i(C_j) r_j, \\ E_{P_1} \sum_{j \in N} E_{P_{2,(j)}} U^i(S_1, S_2) &:= \text{EV}(0, i | \mathbf{P}, \mathbf{P}_{-1}, \dots, \mathbf{P}_{-n}) + \sum_{j \in N} \text{EV}(j, i | \mathbf{P}, \mathbf{P}_{-1}, \dots, \mathbf{P}_{-n}) (1 - f^i(C_j)). \end{aligned}$$

One can derive the closed-form solution for $\text{Rev}^i(S_1, S_2)$ and $U^i(S_1, S_2)$ by setting $P_1(S_1) = 1$ and $P_{2,(e)}(S_2) = 1$ for all options $e \in N$.

EC.4. Extension: Option Substitution under Size Filtering

In this extension, we combine size filtering with option substitution to form a *hybrid* model. Unlike in the option-first revisit model, the retailer observes the customer’s size upon entering the category page, allowing it to tailor both the first- and second-stage assortments accordingly.

Let P_1^i denote the distribution over first-stage assortments offered to customers in size group i , and let $P_{2,(j)}^i$ denote the distribution over second-stage assortments faced by a size- i customer who

selected option j in the first stage. Then, the retailer's optimization problem under size filtering with option substitution is:

$$\begin{aligned}
\Pi_{\text{Hybrid}} = & \max_{\psi, P_1^i, P_{2,(1)}^i, \dots, P_{2,(n)}^i} \sum_{i \in G} \xi^i E_{P_1^i} \sum_{j \in N} E_{P_{2,(j)}^i} \text{Rev}^i(S_1, S_2) - \psi \\
\text{s.t. } & \psi \geq \lambda \left[E_{P_1^k} \sum_{j \in N} E_{P_{2,(j)}^k} U^k(S_1, S_2) - E_{P_1^l} \sum_{j \in N} E_{P_{2,(j)}^l} U^l(S_1, S_2) \right] \quad \forall k, l \in G, \\
& \sum_{S_1 \subseteq N} P_1^i(S_1) = 1 \quad \forall i \in G, \quad \sum_{S_2 \subseteq N} P_{2,(j)}^i(S_2) = 1 \quad \forall j \in N, \forall i \in G, \\
& P_1^i(S_1), P_{2,(j)}^i(S_2) \geq 0 \quad \forall S_1 \subseteq N, S_2 \subseteq N, j \in N, i \in G,
\end{aligned} \tag{HP-Assortment}$$

where (HP-Assortment) stands for *Hybrid Problem - Assortment-Based*.

Similar to the size-filtering and option-substitution models, it is easy to show that (SFP-Assortment) and (RBP-Assortment) are more constrained cases of (HP-Assortment). As such, it is unsurprising that the hybrid model's objective exceeds that of the other two models. In our simulations on 100 randomly generated inventory samples, we observed an average increase of 12.02% in revenue and a 61.03% decrease in unfairness compared to the baseline model with $\lambda = 0.1 \max_{j \in N} r_j$.

We apply the selection-based transformation to find the optimal assortment. Because the retailer can now tailor both the first- and second-stage assortments to a customer's size, we introduce the first-stage selection vectors $\vec{X} := \{X^1, \dots, X^g\}$, where $X^i := \{x_1^i, \dots, x_j^i, \dots, x_n^i\}$, for the first stage, and the set of second-stage selection vectors $\vec{Y} := \{Y^1, \dots, Y^g\}$, where $Y^i := \{y_1^i, \dots, y_n^i\}$ for each $i \in G$. We use index 0 to denote the no-purchase choice in both stages. We introduce a set of vectors of auxiliary variables $\mathcal{Q} := \{\vec{Q}^1, \dots, \vec{Q}^g\}$, where $\vec{Q}^i := \{Q_1^i, \dots, Q_n^i\}$ and $Q_j^i := \{q_{j,0}^i, q_{j,1}^i, \dots, q_{j,n}^i\}$ for each $j \in N$ and $i \in G$.

$$\begin{aligned}
\Pi_{\text{Hybrid}}^{\text{Selection-Based}} := & \max_{\psi, \mathbf{x}, \vec{Y}, \mathbf{u}, \mathcal{Q}} \sum_{i \in G} \xi^i \sum_{j \in N} (x_j^i + y_j^i) f^i(C_j) r_j - \psi \\
\text{s.t. } & \psi \geq \lambda [u^l - u^k] \quad \forall k, l \in G^2, \\
& u^i = (x_0^i + y_0^i) + \sum_{j \in N} (x_j^i + y_j^i) (1 - f^i(C_j)) \quad \forall i \in G, \\
& \frac{x_j^i}{v_j} \leq \frac{x_0^i}{v_0} \quad \forall j \in N, i \in G, \quad \frac{q_{j,m}^i}{v_m} \leq \frac{q_{j,0}^i}{v_0}, \quad j \in N, m \in N, i \in G, \\
& \sum_{j \in N} x_j^i + x_0^i = 1 \quad \forall i \in G, \quad \sum_{m \in N} q_{j,m}^i + q_{j,0}^i = \delta x_j^i, \quad \forall j \in N, i \in G, \\
& y_m^i = \sum_{j \in N} q_{j,m}^i (1 - f^i(C_j)) \quad i \in G, m \in N \cup \{0\}; \\
& q_{m,j}^i \geq 0 \quad \forall m \in N, j \in N \cup \{0\}, \quad q_{j,j}^i = 0 \quad \forall j \in N, \quad x_j^i \geq 0 \quad \forall j \in N \cup \{0\}, i \in G,
\end{aligned} \tag{HP-Selection}$$

where (HP-Selection) stands for *Hybrid Problem - Selection-Based*. We solve this problem and then run a backward transformation similar to that of the revisit behavior model to obtain allocation $\hat{\mathcal{P}}$. We now prove the following result.

THEOREM EC.1. (i): *The optimal values of (HP-Assortment) and (HP-Selection) are equal.* (ii) *Allocation $\hat{\mathcal{P}}$ achieves the optimal solution of (HP-Assortment).* (iii) *Expected revenue and unfairness are identical in the two formulations.*

To save space, we omit the detailed proof of Theorem EC.1. However, by comparing (HP-Selection) and (RBP-Selection), one can easily see their similarities.

EC.5. Extension: Endogenous Configuration Decisions

In the baseline model, we assume that the configuration of each option is exogenous: if an option j is included in an assortment, then all sizes contained in its configuration set C_j are displayed on the option's webpage. We now relax this assumption and extend the model to allow for *endogenous configuration decisions*, where the retailer can choose which subset of sizes to display for each option. To illustrate, suppose option j is available in sizes $\{XS, S, M, XL\}$. Under the baseline model, displaying option j implies displaying all four sizes. Under the endogenized configuration model, the retailer may instead display this option in the configuration $\{XS, XL\}$.

Allowing configuration choices substantially increases modeling complexity because the availability of each SKU affects substitution patterns among neighboring sizes. To manage this complexity, we distinguish between the *full configuration* C_j of option j and its *displayed configuration* $H_j \subseteq C_j$. In addition to deciding which options to include, the retailer now chooses how each option is displayed. We define an *extended assortment* as a tuple $(S, \mathbf{H})$, where $S \subseteq N$ denotes the set of displayed options and $\mathbf{H} := \{H_1, \dots, H_n\}$ specifies the displayed configuration of each option. We also introduce $\mathbf{C} := \{C_1, \dots, C_n\}$ as the set of full configurations of all options. The retailer's problem becomes:

$$\begin{aligned}
 \Pi_{\text{Endogenized}} &= \max_{\mathbf{P}, \psi} \sum_{S, \mathbf{H}} P(S, \mathbf{H}) \text{Rev}(S, \mathbf{H}) - \psi \\
 \text{s.t.} \quad &\psi \geq \lambda \sum_{S, \mathbf{H}} P(S, \mathbf{H}) [U^k(S, \mathbf{H}) - U^l(S, \mathbf{H})] \quad \forall k, l \in G, \\
 &\sum_{S, \mathbf{H}} P(S, \mathbf{H}) = 1, \\
 &P(S, \mathbf{H}) \geq 0 \quad \forall S \subseteq N, \mathbf{H} \subseteq \mathbf{C}, \quad P(S, \mathbf{H}) = 0 \quad \text{if } \mathbf{H} \not\subseteq \mathbf{C},
 \end{aligned} \tag{ECP-Assortment}$$

where (ECP-Assortment) stands for *Endogenized Configuration Problem - Assortment-Based*. In this problem, the constraint $\mathbf{H} \not\subseteq \mathbf{C}$ ensures feasibility by requiring that each displayed configuration satisfies $H_j \subseteq C_j$. We extend the selection-based LP approach to accommodate endogenous configurations. Since the configuration of an option does not affect its selection probability in the

baseline model, we can decouple the probability that customers select an option from the probability that it is seen in a given configuration. We call this the ‘selection-configuration decoupling property’. To see that, let x_j denote the fraction of customers who select option j , and $z_{j,H}$ denote the fraction of customers who select option j and observe it in configuration H .⁵ The resulting linear program is:

$$\begin{aligned}
\Pi_{\text{Endogenized}}^{\text{Selection-Based}} &= \max_{\mathbf{x}, \mathbf{Z}, \psi} \sum_{j \in N} \sum_{H \subseteq C_j} \tilde{r}_j(H) z_{j,H} - \psi \\
\text{s.t.} \quad &\psi \geq \lambda \sum_{j \in N} \sum_{H \subseteq C_j} z_{j,H} (f^k(H) - f^l(H)) \quad \forall k \in G, \forall l \in G, \\
&\frac{x_j}{v_j} \leq \frac{x_0}{v_0} \quad \forall j \in N, \\
&\sum_{j \in N} x_j + x_0 = 1, \quad \sum_{H \subseteq C_j} z_{j,H} = x_j \quad \forall j \in N, \\
&x_j, z_{j,H} \geq 0 \quad \forall j \in N, H \subseteq C_j, \quad z_{j,H} = 0 \quad \forall j \in N, H \not\subseteq C_j,
\end{aligned} \tag{ECP-Selection}$$

where (ECP-Selection) stands for *Endogenized Configuration Problem - Selection-Based*.

After obtaining the optimal transformed variables $\hat{\mathbf{x}}$ and $\hat{\mathbf{Z}}$, we run the backward transformation in two steps. First, by applying the transformation in (13), we obtain the configuration-agnostic display probability $\hat{P}_{CA}$, which is independent of each option’s configuration decision. Then we run an iterative algorithm (which we will discuss shortly) that uses the decoupling property to assign the optimal display probability to each configuration. After running this algorithm, we obtain the optimal display probability $\hat{P}$. Finally, we prove the following theorem.

THEOREM EC.2. (i) *The optimal values of (ECP-Assortment) and (ECP-Selection) are equal.* (ii) *Allocation $\hat{P}$ achieves the optimal solution of (ECP-Assortment).* (iii) *Expected revenue and unfairness are identical in the two formulations.*

The baseline problem is a more constrained version of (ECP-Assortment). Nonetheless, since endogenizing configuration decisions can only lead to hiding an SKU from the display (despite its existence in the inventory), it cannot increase the revenue—unlike the size-filtering or option-substitution extensions. However, especially for larger values of λ , it can drastically reduce unfairness.

⁵ With a slight abuse of notation, we use the vector H as an index in $z_{j,H}$ to keep the notation concise.

Algorithm 1: Assortment-Configuration-Selection

Data: Items $N = \{1, \dots, n\}$; Initial Assortments $\{S_m\}_{m=0}^n$; Initial Display Probabilities $\{\hat{P}(S_m)\}_{m=0}^n$; Configuration Probabilities $\mathbf{Z}$

Result: A list of weighted Assortment-Configuration tuples $\mathcal{A} = \{A_k\}$, where $\sum_k P_k = 1$.

// Initialization: Create a list of tuples $\mathcal{A} = (S, C, P)$

```

1 Initial Assortments  $\mathcal{S} \leftarrow \{S_0, S_1, \dots, S_n\}$ 
2  $\mathcal{A} \leftarrow \emptyset$ 
3 for  $m \leftarrow 1$  to  $n$  do
4    $C_m \leftarrow \mathbf{0}$  // Initialize with empty configurations for all options
5    $P_m \leftarrow \hat{P}(S_m)$  // Assign initial option probabilities
6    $\mathcal{A} \leftarrow \mathcal{A} \cup \{(S_m, C_m, P_m)\}$ 
7 end
// Iterate over options to sequentially assign configurations
8 for  $j \leftarrow 1$  to  $n$  do
9   if  $\hat{P}_j = 0$  then
10    | Break // Skip the option if target probability is 0
11  end
// Sort  $\mathcal{A}$  to prioritize assortments containing  $j$ 
12   $\mathcal{A}_{\text{has},j} \leftarrow \{A_k \in \mathcal{A} : j \in S_k\}$ 
13   $\mathcal{A}_{\text{no},j} \leftarrow \{A_k \in \mathcal{A} : j \notin S_k\}$ 
14   $\mathcal{A} \leftarrow \mathcal{A}_{\text{has},j}$  followed by  $\mathcal{A}_{\text{no},j}$ 
15   $k \leftarrow 1$  // Reset index in the list  $\mathcal{A}$ 
// Iterate over all possible configurations
16  for every  $H \subseteq C_j$  do
17     $p_{\text{remain}} \leftarrow \hat{z}_{j,H}$  // Target marginal probability for  $(j, \text{config } H)$ 
18    while  $p_{\text{remain}} \geq P_k \cdot \Pr(j|S_k)$  do
19       $C_k[j] \leftarrow H$  // Option  $j$  in assortment  $A_k$  is assigned to configuration  $H$ 
20       $p_{\text{remain}} \leftarrow p_{\text{remain}} - P_k \cdot \Pr(j|S_k)$ 
21       $k \leftarrow k + 1$ 
22    end
23    if  $0 < p_{\text{remain}} < P_k \cdot \Pr(j|S_k)$  then
24      // Split the assortment: Create a copy of  $A_k$ , insert it after  $A_k$ ,
25      // and shift indices
26       $A_{\text{new}} \leftarrow \text{Copy}(A_k)$ 
27       $\mathcal{A} \leftarrow \text{Insert}(A_{\text{new}}, \text{after } k)$  // New assortment is at  $k+1$ 
28      // Recalculate probabilities for  $A_k$  and  $A_{k+1}$ 
29       $P_{k+1} \leftarrow P_k - p_{\text{remain}} / \Pr(j|S_k)$ 
30       $P_k \leftarrow p_{\text{remain}} / \Pr(j|S_k)$ 
31       $C_k[j] \leftarrow H$  // Assign configuration  $H$  to option  $j$  in  $A_k$ 
32       $k \leftarrow k+1$  // Move to the next (unassigned) assortment
33    end
34  end
35 end

```

The conversion from the optimal solution for (ECP-Selection) to (ECP-Assortment) happens in two stages. First, we re-index $\hat{x}$ so that $\hat{x}_1/v_1 \geq \dots \geq \hat{x}_n/v_n$. Then, we define S_m and S_0 similar to the baseline model, with $\hat{P}(S_m)$ defined the same way as (13). We define $\hat{P}_j := \sum_{m=j}^n \hat{P}(S_m)$, which shows the probability of displaying an assortment with option j in it. Until this point, we have not assigned configuration display probabilities. We develop Algorithm 1 for that purpose.

Fundamentally, what Algorithm 1 does is ensure that for each option j , each display configuration H gets the display probability $z_{j,H}$. This is feasible because we can *decouple* option and configuration selection decisions. Since we assumed the customer cannot observe the displayed configuration of option j until she chooses that option, x_j and $z_{j,H}$ are connected only through $\sum_{H \subseteq C_j} z_{j,H} = x_j$. Therefore, we first compute the display probabilities for each option, then implement the configuration selection decisions.

The challenge here is managing the display probability of each configuration. Consider the following example. Suppose the optimal solution of the first stage consists of displaying a single assortment with option j in it: $\hat{P}(S_1 = \{1, \dots, j, \dots, m\}) = 1$. However, only for option j , we have $\hat{z}_{j,H_1}/\hat{x}_j = 0.5$ and $\hat{z}_{j,H_2}/\hat{x}_j = 0.5$, so we must ensure that half of the customers clicking on option j must see it in configuration H_1 , and the other half must see it in configuration H_2 . To do so, we conduct *probability-preserving assortment splitting* (or assortment splitting, for short): we make a copy (i.e., everything the same) of assortment S_1 , call it assortment S_2 , and set the configuration of option j in S_1 as H_1 , and in S_2 as H_2 . Then, we randomize over S_1 and S_2 with equal probability. First, note that the selection probability of option j before and after the split remains unchanged, because the options in S_1 and S_2 are the same. Second, now half of the customers choosing option j see it in configuration H_1 , and the other half must see it in configuration H_2 .

We now prove Theorem EC.2. Suppose $(\hat{\mathbf{x}}, \hat{\mathbf{Z}}, \hat{\psi})$ is the optimal solution for (ECP-Selection). We first get $\hat{\mathbf{P}}$ using (13) and then run Algorithm 1 to get $\mathcal{A}$. We define the list of weighted assortment-configuration tuples after the j -th iteration with $\mathcal{A}^j$. We have the following lemma.

LEMMA EC.1. *After running Algorithm 1, we have:*

1. $\sum_{(S, \mathbf{C}, P) \in \mathcal{A}^n} P(S, \mathbf{C}) \Pr(j|S) = \hat{x}_j$ for any option j ;
2. $\sum_{(S, \mathbf{C}, P) \in \mathcal{A}^n} P(S, \mathbf{C}) \Pr(j|S) \cdot \mathbf{1}(\mathbf{C}_j = H) = \hat{z}_{j,H}$ for any option-configuration pair (j, H) .

Proof of Lemma EC.1 We prove this lemma using induction. Define $\mathcal{B}_{\text{has},m}^j \leftarrow \{A_k \in \mathcal{A}^j : m \in S_k\}$ and $\mathcal{D}_{\text{has},m,H}^j \leftarrow \{A_k \in \mathcal{A}^j : m \in S_k, C_m = H\}$. Before the first iteration, we have

$$\sum_{(S, \mathbf{C}, P) \in \mathcal{B}_{\text{has},j}^0} P(S, \mathbf{C}) \Pr(j|S) = \sum_{m=j}^n \hat{P}(S_j) \Pr(j|S) = \sum_{m=j}^n \left(\frac{\hat{x}_m}{v_m} - \frac{\hat{x}_{m+1}}{v_{m+1}} \right) \left[v_0 + \sum_{i \in S_m} v_i \right] \frac{v_j}{v_0 + \sum_{i \in S_m} v_i} = \hat{x}_j.$$

Similarly, we have $\sum_{(S, \mathbf{C}, P) \in \mathcal{D}_{\text{has},j,0}^0} P(S, \mathbf{C}) \Pr(j|S) \cdot \mathbf{1}(C_j = \mathbf{0}) = \hat{x}_j$, and for any $H \neq \mathbf{0}$ we have $\sum_{(S, \mathbf{C}, P) \in \mathcal{D}_{\text{has},j,q}^0} P(S, \mathbf{C}) \Pr(j|S) \cdot \mathbf{1}(C_j = H) = 0$.

When running the first iteration, we divide our assortment into two mutually exclusive subsets $\mathcal{B}_{\text{has},1}^0$ and $\mathcal{A}^0 \setminus \mathcal{B}_{\text{has},1}^0$. Assortments in $\mathcal{A}^0 \setminus \mathcal{B}_{\text{has},1}^0$ are not affected by the first iteration. That is, $\mathcal{A}^0 \setminus \mathcal{B}_{\text{has},1}^0 = \mathcal{A}^1 \setminus \mathcal{B}_{\text{has},1}^1$. However, the same cannot be said for assortments in $\mathcal{B}_{\text{has},1}^0$ because of assortment modification and/or assortment splitting during the first iteration.

Suppose during the first iteration, Algorithm 1 modifies assortment A_k . We know the modification process during the first iteration does not affect the configurations of any option other than option 1. As such, we have that $P_k \Pr(j|S_k) \mathbf{1}(C_k[j] = H)$ remains unchanged. Now, suppose that during the first iteration, Algorithm 1 splits assortment A_l and produces a modified assortment A_l and a new assortment A_{l+1} . Because A_{l+1} is a copy of A_l , we have $\Pr(j|S_l) \mathbf{1}(C_l[j] = H) = \Pr(j|S_{l+1}) \mathbf{1}(C_{l+1}[j] = H)$. Also, we have that $P_l + P_{l+1}$ after splitting equals P_l before splitting. That means $P_l \Pr(j|S_l) \mathbf{1}(C_l[j] = H) + P_{l+1} \Pr(j|S_{l+1}) \mathbf{1}(C_{l+1}[j] = H)$ after splitting and $P_l \Pr(j|S_l) \mathbf{1}(C_l[j] = H)$ before splitting are equal. Therefore, for any configuration H and any option except option 1, we have $\sum_{(S,C,P) \in \mathcal{B}_{has,1}^1} P(S,C) \Pr(j|S) \cdot \mathbf{1}(C_j = H) = \sum_{(S,C,P) \in \mathcal{B}_{has,1}^0} P(S,C) \Pr(j|S) \cdot \mathbf{1}(C_j = H)$. Using that, we have $\sum_{(S,C,P) \in \mathcal{B}_{has,1}^1} P(S,C) \Pr(j|S) = \sum_{(S,C,P) \in \mathcal{B}_{has,1}^0} P(S,C) \Pr(j|S)$. Combining the two types of assortments, for any option-configuration pair (j, H) with $j \neq 1$, we have $\sum_{(S,C,P) \in \mathcal{A}^1} P(S,C) \Pr(j|S) \cdot \mathbf{1}(C_j = H) = \sum_{(S,C,P) \in \mathcal{A}^0} P(S,C) \Pr(j|S) \cdot \mathbf{1}(C_j = H)$, which in turn means $\sum_{(S,C,P) \in \mathcal{A}^1} P(S,C) \Pr(j|S) = \sum_{(S,C,P) \in \mathcal{A}^0} P(S,C) \Pr(j|S) = \hat{x}_j$.

We now explain the changes toward option 1 in the first iteration. For this, consider configuration H . Without loss of generality, suppose through modification and/or splitting, Algorithm (1) assigns configuration H to assortments A_1 to A_m . From the way that p_{remain} is updated in the algorithm, we know that $\sum_{k=1}^m P_k \Pr(1|S_k) = \hat{z}_{1,H}$. Therefore, $\sum_{(S,C,P) \in \mathcal{A}^1} P(S,C) \Pr(1|S) \cdot \mathbf{1}(C_1 = H) = \hat{z}_{1,H}$. Also, from the relevant constraint in (ECP-Selection), we know that

$$\sum_{(S,C,P) \in \mathcal{A}^1} P(S,C) \Pr(1|S) = \sum_{H \subseteq C_j} \sum_{(S,C,P) \in \mathcal{A}^1} P(S,C) \Pr(1|S) \cdot \mathbf{1}(C_1 = H) = \sum_{H \subseteq C_j} \hat{z}_{1,H} = \hat{x}_1.$$

Therefore, $\sum_{(S,C,P) \in \mathcal{A}^1} P(S,C) \Pr(j|S) = \hat{x}_j$ for any option j , and $\sum_{(S,C,P) \in \mathcal{A}^1} P(S,C) \Pr(1|S) \cdot \mathbf{1}(C_1 = H) = \hat{z}_{1,H}$ for any configuration H . This concludes the first step of induction.

Suppose $\sum_{(S,C,P) \in \mathcal{A}^k} P(S,C) \Pr(j|S) = \hat{x}_j$ for any option j , and $\sum_{(S,C,P) \in \mathcal{A}^k} P(S,C) \Pr(j|S) \cdot \mathbf{1}(C_j = H) = \hat{z}_{j,H}$ for all option-configuration pairs (j, H) where $j \leq k$. We run the $k+1$ -th iteration. The proof steps are the same as those we followed for the first iteration. We first divide $\mathcal{A}^{k+1}$ into two groups: $\mathcal{B}_{has,k+1}^{k+1}$ and $\mathcal{A}^{k+1} \setminus \mathcal{B}_{has,k+1}^{k+1}$. We know that the algorithm does not change the configuration of any option other than option $k+1$ during the $k+1$ -th iteration. Therefore, for any option-configuration pair (j, H) with $j \neq k+1$, we have $\sum_{(S,C,P) \in \mathcal{B}_{has,k+1}^{k+1}} P(S,C) \Pr(j|S) \cdot \mathbf{1}(C_j = H) = \sum_{(S,C,P) \in \mathcal{B}_{has,k+1}^k} P(S,C) \Pr(j|S) \cdot \mathbf{1}(C_j = H)$. Using that, we have $\sum_{(S,C,P) \in \mathcal{B}_{has,k+1}^{k+1}} P(S,C) \Pr(j|S) = \sum_{(S,C,P) \in \mathcal{B}_{has,k+1}^k} P(S,C) \Pr(j|S)$. Combining the two types of assortments, for any option-configuration pairs (j, H) with $j \neq k+1$ we have $\sum_{(S,C,P) \in \mathcal{A}^{k+1}} P(S,C) \Pr(j|S) \cdot \mathbf{1}(C_j = H) = \sum_{(S,C,P) \in \mathcal{A}^k} P(S,C) \Pr(j|S) \cdot \mathbf{1}(C_j = H)$, which in turn means $\sum_{(S,C,P) \in \mathcal{A}^{k+1}} P(S,C) \Pr(j|S) = \sum_{(S,C,P) \in \mathcal{A}^k} P(S,C) \Pr(j|S) = \hat{x}_j$. Also,

from the induction assumption, we see that $\sum_{(S, \mathbf{C}, P) \in \mathcal{A}^{k+1}} P(S, \mathbf{C}) \Pr(j|S) \cdot \mathbf{1}(C_j = H) = \sum_{(S, \mathbf{C}, P) \in \mathcal{A}^k} P(S, \mathbf{C}) \Pr(j|S) \cdot \mathbf{1}(C_j = H) = \hat{z}_{j,H}$ for any $j \leq k$. We then follow the same steps as for option 1 to prove that $\sum_{(S, \mathbf{C}, P) \in \mathcal{A}^{k+1}} P(S, \mathbf{C}) \Pr(j|S) \cdot \mathbf{1}(C_{k+1} = H) = \hat{z}_{k+1,H}$, so $\sum_{(S, \mathbf{C}, P) \in \mathcal{A}^{k+1}} P(S, \mathbf{C}) \Pr(k+1|S) = \hat{x}_{k+1}$ after running $k+1$ -th iteration, concluding the proof. $\square$

With Lemma EC.1, proving that the optimal solution for (ECP-Selection) can be converted into a feasible solution for (ECP-Assortment) with the same revenue and unfairness terms is straightforward and follows the same steps as in the baseline model. For the reverse relation, suppose $\mathbf{P}'$ is the optimal solution for (ECP-Assortment). Define:

$$x'_j := \sum_{P \in \mathbf{P}'} \sum_H P(S, \mathbf{C}) \cdot \Pr(j|S) \cdot \mathbf{1}(C_j = H), \quad z'_{j,H} := \sum_{P \in \mathbf{P}'} P(S, \mathbf{C}) \cdot \Pr(j|S) \cdot \mathbf{1}(C_j = H) \quad \forall j \in N, H \subseteq G.$$

Using these relations, it is straightforward to show that an optimal solution of (ECP-Assortment) maps to a feasible solution of (ECP-Selection) with the same revenue and unfairness. Combining the two directions, we prove the equivalence of the two programs.

EC.6. Model Calibration

In this section, we apply a Difference-in-Differences (DiD) method to estimate the model coefficients using a granular, SKU-level dataset we acquired in collaboration with an industry partner. For estimation and calibration, we normalize the outside option attraction v_0 to be equal to one.

EC.6.1. The Dataset

Our industry partner is a major European fast-fashion retailer that provided us with access to SKU-level sales data from its online store over a two-month period. We focus on the women's long-sleeve shirts category in this study. On average, 238 distinct options and 970 distinct SKUs were displayed daily on this category's page. The average daily sales for this category are 80, which indicates an average sales-per-SKU of 0.082.

EC.6.2. DiD Analysis and the Structural Model

To estimate the values of p^u and p^l in (2), we adopt the DiD approach in Li et al. (2023). The level of granularity in our data allows us to adopt the DiD method in Li et al. (2023), which does not require complex estimation methods or assumptions on latent variables (such as the methods employed in, e.g., Vulcano et al. (2012) and Musalem et al. (2010)).

The key idea in Li et al. (2023) is to focus on the SKU-level data and to choose the control and treatment groups from different sizes of the same option. This approach eliminates factors common across different option sizes and focuses on the effects of differences in size groups. This is particularly relevant to the in-store shoe-shopping experience they study, as all customers who

visit a store on a given day see the same assortment of shoes. After choosing the shoes (brand, design, color, etc.) they prefer, customers ask for them in their size. So, to the point of checking whether an option is available in their size, customers of all sizes have similar experiences. Because our baseline choice model assumes the same, we adopt their method to estimate the parameters of interest.

Suppose on the day t , N_t customers (in all sizes) visit the online store with the assortment (S_t) . The expected number of customers who visit the page for option j on that day is $N_t \frac{v_j}{1 + \sum_{k \in S_t} v_k}$, which is independent of the configuration. Sales of this option (i.e., SKUs that belong to it) depend on its configuration. More specifically, the expected sales of size i of option j are:

$$E[\text{Sales}_{jt}^i | S_t, \mathbf{C}_t, N_t] = N_t \frac{v_j}{1 + \sum_{k \in S_t} v_k} \mathbf{1}_{(i \in C_{jt})} (\xi^i + p^u \mathbf{1}_{(i-1 \notin C_{jt})} \xi^{i-1} + p^l \mathbf{1}_{(i+1 \notin C_{jt})} \xi^{i+1}), \quad (\text{EC.11})$$

where $\mathbf{C}_t := \{C_{1t}, \dots, C_{nt}\}$. In (EC.11), an SKU's sales depend not only on its availability but also on the (un)availability of its neighboring sizes. This helps us to choose the treatment and control groups similar to Li et al. (2023). The following example elaborates on the approach.

Suppose the available size groups are $G = \{\text{XS}, \text{S}, \text{M}, \text{L}, \text{XL}, \text{XXL}\}$, and on day 1, $C_{j1} = G$, i.e., all the sizes of option j are available and so size substitution for this option is not a concern. But on day 2, $C_{j2} = G \setminus \{\text{S}\}$, so all sizes for option j are available except size S. We use equation (EC.11) to write the expected sales of sizes M and L over two days in Table EC.1:

	Day 1	Day 2
$E[\text{Sales}_j^M S, \mathbf{C}, N]$	$N_1 \frac{v_j}{1 + \sum_{k \in S_1} v_k} \xi^M$	$N_2 \frac{v_j}{1 + \sum_{k \in S_2} v_k} (\xi^M + \xi^S p^u)$
$E[\text{Sales}_j^L S, \mathbf{C}, N]$	$N_1 \frac{v_j}{1 + \sum_{k \in S_1} v_k} \xi^L$	$N_2 \frac{v_j}{1 + \sum_{k \in S_2} v_k} \xi^L$

Table EC.1 Expected sales of option j for SKUs M and L according to (EC.11).

Table EC.1 provides the structure of the DiD estimations. We divide the shocks affecting an option's sales into two groups. The first group consists of factors that depend on option assortment on day t (i.e., S_t), and the second group encompasses factors depending on the configuration of option j on that day (i.e., C_{jt}). Both M and L sizes follow *parallel trends* in the sense that changes in S_t affect both similarly. However, by equation (EC.11), unavailability in size S of option j affects the expected sales of size M but not the sales of size L. In other words, had it not been for the unavailability of size S, sizes M and L of option j would have followed a parallel trend.

Assuming that N_t follows a Poisson distribution (see e.g., Vulcano et al. (2012) and Li et al. (2023)), it must be that daily sales count follows a Poisson distribution, i.e., $\text{Sales}_{jt}^i | S_t, \mathbf{C}_t \sim$

Poisson(ϕ_{jt}^i). Therefore, using (EC.11), for the days where $j \in S_t$ and $i \in C_{jt}$, we have:

$$\begin{aligned} \log(\phi_{jt}^i) &= \log\{E[\text{Sales}_{jt}^i | S_t, C_t]\} = \log\left(\frac{N_t}{1 + \sum_{k \in S_t} v_k}\right) + \log(v_j) + \log(\xi^i) \\ &\quad + \log\left(1 + p^u \mathbf{1}_{(i-1 \notin C_{jt})} \frac{\xi^{i-1}}{\xi^i} + p^l \mathbf{1}_{(i+1 \notin C_{jt})} \frac{\xi^{i+1}}{\xi^i}\right) \\ &= \alpha_t + \beta_j + \zeta_i + SSE_{jt}^i, \end{aligned} \quad (\text{EC.12})$$

where $SSE_{jt}^i = \log\left(1 + p^u \mathbf{1}_{(i-1 \notin C_{jt})} \xi^{i-1}/\xi^i + p^l \mathbf{1}_{(i+1 \notin C_{jt})} \xi^{i+1}/\xi^i\right)$ captures the size substitution effect. The last equality in (EC.12) holds because the first term on the right-hand side is only a function of t , the second term is only a function of the option j 's perceived value, and the third term is only a function of the fraction of people in size-group i .

As SSE_{jt}^i is nonlinear, we use a linear approximation. The two parameters p^u and p^l are unlikely to be large numbers and so we apply the first-order Taylor expansion of $\log(1+x)$ to write: $SSE_{jt}^i \cong p^u \mathbf{1}_{(i-1 \notin C_{jt})} \xi^{i-1}/\xi^i + p^l \mathbf{1}_{(i+1 \notin C_{jt})} \xi^{i+1}/\xi^i$. This helps rewrite equation (EC.12) as

$$\log(\phi_{jt}^i) \cong \alpha_t + \beta_j + \zeta_i + p^u \mathbf{1}_{(i-1 \notin C_{jt})} \frac{\xi^{i-1}}{\xi^i} + p^l \mathbf{1}_{(i+1 \notin C_{jt})} \frac{\xi^{i+1}}{\xi^i}. \quad (\text{EC.13})$$

To estimate (EC.13), we define six scenarios. In Scenario 2, we estimate the size substitution effect of unavailability in size S on size M. To do so, we track four sizes of option j : S, M, L, and XL. We only focus on the days when sizes M, L, and XL are available—size S may or may not be present. Similar to the example in Table EC.1, the sales for size L of option j should not be affected by the absence or presence of size S, as both of its neighbors are available to the customers. This explains why size XL needs to be in the assortment as well. For size M, on the other hand, we expect to observe the size substitution effect from size S to size M on days when size S is absent from the assortment. Therefore, in Scenario 2, we choose size M as our *treatment* group and size L as our *control* group. With the same logic, we introduce the remaining five scenarios in Table EC.2. The first three scenarios occur when a larger size is substituted for an unavailable size, while the last three scenarios occur when a smaller size is substituted.

Scenario	Out-of-Stock(o)	Treatment(t)	Control(c)	Neighbor(n)
1	XS	S	M	L
2	S	M	L	XL
3	M	L	XL	XXL
4	L	M	S	XS
5	XL	L	M	S
6	XXL	XL	L	M

Table EC.2 Scenarios in the DiD Regressions.

Based on these scenarios, we create a DiD regression for the focal category. Obviously, for identification in each scenario, we focus on options for which there were days when they were available in all four sizes and days when they were unavailable in a size.

Given the setup of the DiD model, a treatment size cannot be the target of the substitution effect by two neighboring sizes at the same time. So if an option is used for a scenario, we do not use it for another ‘conflicting scenario’ (e.g., for Scenarios 2 and 3 where the treatment and control sets overlap). Overall:

$$\begin{aligned} \log(\phi_{jt}^i) = & \pi + p \times \frac{\xi^o}{\xi^t} \times \text{Treat}_{jt}^{io} \times \text{OOS}_{jt}^o + \kappa \times \text{Treat}_{jt}^{io} + \theta \times \text{OOS}_{jt}^o + \phi \log(\text{price}_{jt}^i) \\ & + \alpha_t + \beta_j + \zeta_i + \varepsilon_{jti}. \end{aligned} \quad (\text{EC.14})$$

The variable p is the size substitution coefficient to be estimated; Treat_{jt}^{io} is equal to 1 if, in the scenario under discussion, size i is the neighbor of the out-of-stock size ‘ o ’ and 0 otherwise; OOS_{jt}^o is equal to 1 if size ‘ o ’ of option j is out of stock on day t and 0 otherwise; ϕ is the effect of $\log(\text{price}_{jt}^i)$ (we add this term to control for changes in prices during the study period); and π is the constant term. We do not include the inventory level of an SKU as this information is not available to the customers.

Independent Variables	Women Shirt LS		
	All	Sc 1, 2, 3	Sc 4, 5, 6
$\text{Treat}_{jt}^{io} \times \text{OOS}_{jt}^o$	0.2349** (0.1171)	0.0643 (0.2941)	0.2568** (0.1283)
$\log(\text{price})$	-5.3391** (2.1514)	-0.4689 (5.0332)	-5.1724** (2.1535)
Date FE	YES	YES	YES
Option FE	YES	YES	YES
Size FE	YES	YES	YES
Observations	12128	2574	9554
- Log-Likelihood	3761	666	3044

Table EC.3 Regression results for ‘Women Shirts Long Sleeve’. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

We run these regressions in three parts. First, we pool all the samples and assume $p^u \approx p^l$ to obtain a single estimation. To estimate p^u , we run the regression model on samples from Scenarios 1–3 in which size substitution happens with one size larger. The last regression is on samples from Scenarios 4–6 to estimate p^l . Table EC.3 presents the results of the DiD regression on the target category. Due to data limitations, the results for Scenarios 1–3 are insignificant. However, the other two regressions indicate a statistically significant substitution effect. In the pooled specification that enforces $p^u = p^l$, the estimated coefficient is 0.235; in the baseline simulations, we round this to 0.25. This estimate is close to the one reported in Li et al. (2023).

EC.6.3. Simulation Setup

We follow a number of steps to characterize the distribution of C_j ’s (i.e., available sizes in configurations) from the dataset. First, we create a new coding ‘option-date-code’ to assign a unique

number to each option (and its SKUs) for each day. Next, we observe the availability in different sizes for each option-date pair. We set $z_{jt}^i = 1$ if option j is available (has at least one SKU in the inventory) in size i on day t . If an option is absent from the assortment on a day, we do not define z_{jt}^i for that option-date pair. If option j is not produced in size i , we also set $z_{jt}^i = 0$ for every day that option j is present in the assortment. Finally, for each size i , we compute $1 - \bar{z}_i$ across all option-date pairs; the resulting unavailability rates are reported in Table 1. In the simulations, we set the unavailability probability of an option in each size to follow a Bernoulli distribution with parameter p as in Table 1 and assume the availability of different sizes is independent of each other.

For customer size distribution, we use the estimated Ξ for the women's shirts category in Table 1. To estimate Ξ , we focused on option-day pairs where all sizes are available, and compared the sales in different sizes. Table 1 reveals the systematic disparity that we want to alleviate: although size groups *XXL* and *L* represent the same share of customers, the unavailability rate for size group *XXL* is more than twice that of size group *L*.

The next step is choosing the distribution of preference weights ($\{v_j\}$) and revenues ($\{r_j\}$). While prices are available in our dataset, the preference weights must be inferred from the data. By (EC.12), the option-level fixed effects (β_j 's) in regressions represent $\log(v_j)$. Since we set $v_j = \exp(u_j)$ in Section 3, it is the case that $u_j = \log(v_j) = \beta_j$. Therefore, after performing the DiD regressions, we set $\hat{u}_j = \hat{\beta}_j$, where the hat sign shows the estimation results. Figure EC.2 plots $\hat{u}_j$ against r_j for each option as well as their histograms.

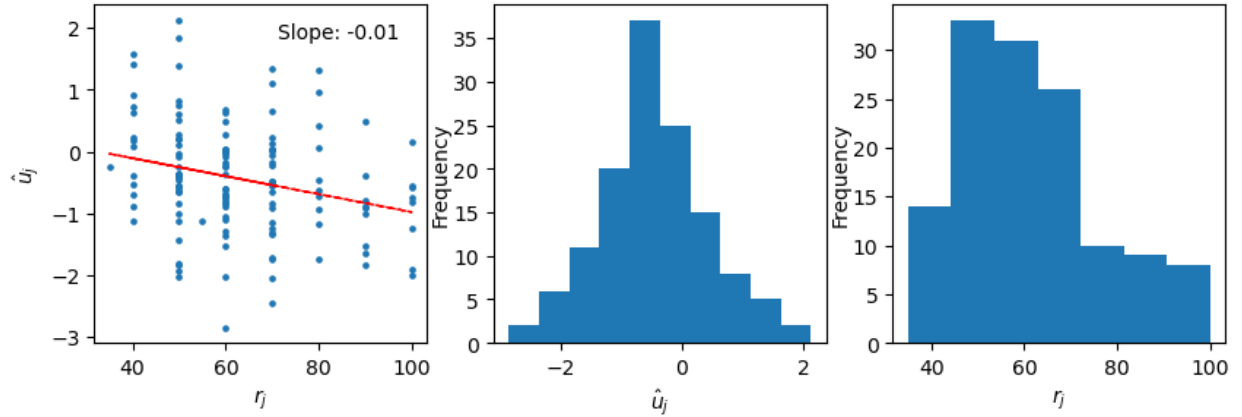


Figure EC.2 $\{r_j\}$ and $\{\hat{u}_j\}$ distribution for women's shirts category.

We make multiple observations from Figure EC.2. First, and typical for a fast fashion retailer, prices are distributed according to a unimodal and right-skewed distribution. Second, $\hat{u}_j$'s tend to follow a unimodal and symmetric distribution. Third, a negative correlation exists between r_j 's and $\hat{u}_j$'s.

We use a Gamma distribution with shape = 10 and scale = 5 to obtain price samples. To mimic the price distribution of the focal category, we assume prices cannot be less than 25 or more than

100. As such, if a price is higher than 100 or less than 25, we replace it with a sample from a uniform distribution between 25 and 100. (For a Gamma distribution with shape = 10 and scale = 5, the mean is 50. Considering the mean of the replaced samples is $62.5 = (100 + 25)/2$, the mean of r_j in our simulations is slightly above 50.)

Next, we obtain a_j as samples of a standard Normal distribution and then build $u_j = a_j - \nu r_j$ with $\nu = 0.02$ to represent the slope. Finally, we set $v_j = \exp(u_j)$ to get the preference weights used in the equations. We do not have access to the number of customers who visited the store on a specific day, so we do not know how many customers left the store without choosing anything. Similar to Sumida et al. (2021), we scale preference weights so that $1/(1 + \sum_{j \in [n]} v_j)$ (the probability that a customer leaves the store without choosing any option if all the options are introduced) is equal to an exogenously set value p_0 . In the simulations, we set $p_0 = 0.25$.